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A Narrative Synthesis Systematic Review of Pakistani Women's Health Outcomes from Primary Level Interventions

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ABSTRACT

Objective

Women living in Pakistan have complex health problems including infectious and non-communicable diseases, accident and injuries, and mental health problems. While a majority of these women rely on primary healthcare services for all of their healthcare needs there has to date been no overview of the extent of their effectiveness. The objective of this review was to (i) synthesise the available evidence regarding the effectiveness of primary care based interventions aimed at improving women's mental and physical health; and (ii) identify the factors that promote effective working.

Methods

Five academic databases were searched, including PubMed, BMC Medicine, Medline, CINAHL and the Cochrane Library. A search was also made of the Grey literature. The quality of included studies was assessed using a standardized critical appraisal tool, and the findings summarized using a narrative synthesis.

Results

In total 18 studies were included in the review. Eight involved evaluations of counseling interventions, three health education and awareness interventions, two social and psychosocial interventions, and five were evaluations of combination interventions. Twelve of the included studies were RCTs. Of these 14 reported significant outcomes, and 4 further interventions showed partially favorable results. However, interventions mostly targeted women's mental or reproductive health.

Conclusions: While the evidence is limited in terms of quality and what has been evaluated, a number of interventions appear to be effective in improving outcomes for women. The three key approaches include the adoption of an active door-to-door and group-based approach; utilization of community peers who can deliver care cost-effectively and who are more accepted in the community; and the integration of financial vouchers to support uptake in poor populations.

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Keywords: Narrative synthesis; systematic review; Pakistan; Women's Health Outcomes; Primary Level Interventions

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Strengths and Limitations of this study

- This is the first synthesis of evidence about the effectiveness of primary-care based interventions for women in Pakistan.
- The quality of the included studies was appraised using a standardized tool.
- The diverse nature of the interventions included in this review and assessment of different outcomes meant that it was only possible to provide a narrative synthesis.
- The small number of studies, limited geographical reach, and use of patient-reported outcomes limits the generalizability of the findings.

Introduction

A majority of Pakistani women are dependent on primary healthcare services that are provided by the government 1 as a result of poverty 2 and informal sector employment, which excludes women from employer health protection 3. Despite high dependence on state provision, primary health care services for women are a neglected area in Pakistan. Planning and services are both crippled by patriarchal traditions, low prioritization, and insufficient funding 4. Women in the country currently face diverse challenges related to high maternal mortality 5, chronic disease 6, infectious diseases 7, a high incidence of injury and violence 8, mental health problems 9, and health issues related to aging 10. The country also has a number of unmet Millennium Development Goals, with key issues such as neonatal and maternal mortality rates remaining relatively unchanged since 2012 11.

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Primary healthcare in Pakistan

Since 2011 responsibility for the delivery of Pakistan's healthcare system has been devolved to the provinces. The Federal Ministry of National Health Service Coordination and Regulation is responsible for monitoring and allocation of the budget to the four provinces of the country including Baluchistan, Khyber Pakhtunkhwa, Sindh and Punjab. The overall framework and structure for the delivery of primary healthcare services is the same across the country, with services including: reproductive and maternal health; vaccination and immunisation; health awareness; referral to secondary and tertiary sectors; and routine counseling. However, the outreach and implementation varies according to provincial development, with Punjab and KPK provinces performing better, compared to Baluchistan, Sindh and Gilgit-Baltistan 12.

Services to women are provided by lady health workers (LHWs), who can be found at the community basic health units (BHUs) and the rural health centres (RHCs), but also make door-to-door visits. The Lady health Workers (LHW) Programme, which was launched by the government of Pakistan in 1994 13 provides key services for immunization, family planning, counseling and maternal health services. The BHUs and RHCs are also allocated a physician or medical officer and nurse, who remain at the centre to support clients for clinical consultancy and basic reproductive health services. Home visits are also made by community midwives (CMWs) to support the delivery of reproductive health services including safe childbirth.

Existing evidence

While a number of studies have reported that the LHW programme has been successful in improving family planning 14 15 and mortality rates 16, others have reported that primary health care services have not yielded positive outcomes for women's health 11 17 18. To date there has

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been no systematic review of rigorous evaluations to identify which primary care interventions are effective in improving women's health outcomes in Pakistan. The aim of this review was to systematically review the available literature for studies that have evaluated the effectiveness of primary healthcare services for women, in order to inform future health research and policy.

METHODS

Patient and public involvement

No patients are involved in this study.

Search strategy

A systematic review using a narrative synthesis was undertaken. Electronic databases including Pubmed, BMC, Medline, CINAHL, and Cochrane Library were searched between the dates August 2020 to October 2020. A search of the grey literature included the following databases: Google, Google Scholar, WHO website and Government of Pakistan Health Ministry websites. Detailed information about the search strategy, search terms, inclusion and exclusion criterion, and methods for this study can be found in the protocol publication 19. A summary of the search terms and database search results can be found in **Supplementary File 1.**

The inclusion criteria were established using PICOS: (i) Population: The population of interest was all Pakistani women, above 18 years of age, and excluding children and adolescent girls; (ii) Intervention: all healthcare interventions that are delivered in the primary healthcare sector; (iii) Comparator: the review only included studies that used a control or comparison group; (iv) Outcomes: All studies that focused on physical or mental health outcomes, were included in the review; (v) Study design: all RCTs and quasi-experimental study designs that involved the use of a control group. The language was restricted to English.

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Initial Screening

An initial screening of studies based on titles, was performed by the lead investigator for this study. In the second screening, titles and abstracts were evaluated by a second reviewer independently.

Data extraction

The following information was abstracted from the interventions that met our eligibility criteria in terms of targeting women's health outcomes: study design; setting; participant type; sample size; aim; type of intervention; outcomes being measured; results.

Critical appraisal

The intervention-based studies were appraised using a standardized critical appraisal tool for RCTs 20. The quality of the articles was assessed by two reviewers, and group meetings were held to discuss the independent appraisal of articles and to resolve discrepancies. More information related to the RCT Critical Appraisal for the included articles in this review has been summarized in Supplementary File 2. Twelve of the eighteen included articles scored high on the six evaluation areas of the critical appraisal checklist. Six studies were rated as medium quality mostly due to the failure to use randomization.

Data management and synthesis

Studies from databases were transferred into Endnote Software. After the exclusion of duplicates, studies were transferred to an Excel template for data extraction and organization. After the final selection of studies to be included, separate tabs were created for each study for detailed data extraction and summary sheets were also created to organize study characteristics and information for supplementary data files. As a result of the high level of heterogeneity across

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studies in terms of interventions and outcomes, a narrative summary of the findings has been presented.

RESULTS

Included Studies

The search results have been reported using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses flow chart, presented in **Figure 1**. A total of 135,553 publications were identified, of which 5,405 were duplicates. After duplicates were removed 109,841 were selected for further screening of titles. Following a review of the abstracts, the full text of 543 publications were retrieved, and assessed against the selection criteria, and a total of 18 studies were included of interventions assessing the effectiveness of primary healthcare interventions for women's health outcomes. **Table 1** lists the interventions including information about the authors, study type, sample, description of intervention, and post-intervention outcomes with p-values. Further details of the characteristics of included studies have been reported in **Supplementary File 3**.

Study characteristics

The interventions included in this review (**Table 1**) were classified as follows: (i) Eight studies were classified as 'Counseling interventions' 15 21 22 27-32; (ii) three as 'Health Education and Awareness Interventions' 23 33-35; (iii) two as 'Social and Psychosocial Interventions' 36-38; and (iv) five as 'Combination Interventions (e.g. *vouchers, education, referral and technology)* 24 25 39-43.

With regard to the study designs: (a) 12 studies are RCTs (including 7 RCTs and 5 cluster RCTs); and (b) 6 are quasi-experimental studies including 1 non-randomized non-inferiority trial. A majority of the studies (n=9) delivered interventions to pregnant women. Another 5

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studies delivered interventions to women of reproductive years. Two studies targeted women aged 18 and above affected by conflict and another 2 delivered the intervention to women screened for depression. Overall therefore, a majority of included studies targeted health outcomes for pregnant women or women of reproductive years.

Nearly all the interventions (n=15) used patient-reported outcomes, with 3 reporting both patient-reported and clinical outcomes. None of the interventions included a sample from across all the provinces of Pakistan. Six of the interventions were limited to only one city or town, 10 to one province of Pakistan, and 2 studies included samples from two or three provinces in the country. A small number of interventions were delivered exclusively by lady health workers (LHWs) (n=5), and another 6 interventions were facilitated by a combination of LHWs, community health workers (CHWs), community midwives (CMWs), traditional birth attendants, and laywomen. The remaining interventions were delivered by either laywomen or local peers (n=3); trained healthcare staff, nurses or health educators (n=3); or by CHWs alone (n=1). Fourteen of the included studies showed significant results for primary outcomes and 4 showed partially favorable results.

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Table 1:
List of included intervention-based studies to review effectiveness on women's health outcomes

Authors	Study Type	Sample	Intervention	Results (p value)
Ali, B. S., Rahbar, M. H., Naeem, S., Gul, A., Mubeen, S., & Iqbal, A. (2003)	RCT	Women aged 18 to 50 years screened for anxiety or depression (Experiment: n= 70; Control: n= 91)	One-on-one counseling sessions to reduce anxiety and/or depression	A significant reduction was found between the mean anxiety and depression scores of both the intervention and control groups (p = 0.000)
Khan, M. N., Hamdani, S. U., Chiumento, A., Dawson, K., Bryant, R. A., Sijbrandij, M., Rahman, A. (2017)	Cluster RCT	Women aged 18 or more years affected by conflict (Experiment: n= 59 Control: n= 60)	Group psychotherapeutic intervention to reduce anxiety and depression in women	Statistically significant decline in PTSD symptoms (p=<0.0001) and depressive disorder scores (p= 0.0016) of the intervention compared to the control group
Rahman, A., Khan, M. N., Hamdani, S. U., Chiumento, A., Akhtar, P., Nazir, H., van Ommeren, M. (2019)	Cluster RCT	Women aged 18–60 years affected by conflict (Experiment: n= 288 Control: n= 290)	Group psychological intervention to improve mental health	A significant lower mean total score of anxiety and depression in women in the intervention group compared to the control group (p=0·0007)
Ali, N. S., Ali, B. S., Azam, I. S., & Khuwaja, A. K. (2010)	Non-randomized quasi-experiment	Mothers, with children 0-30 months, screened for postnatal depression (Experiment: n= 62 Control: n= 12)	One-on-one counseling to reduce postnatal depression	A significant decline in level of anxiety/depression was found in both the counseled and the non-counseled groups (p= < 0.001); with a greater decline in the counseled/ intervention group. The counseled group fared better than the non-counseled for (i) recovery; (ii) reduction in the rate of recurrence; and (iii) increase in the duration before relapse.
Sikander, S., Ahmad, I., Atif, N., Zaidi, A., Vanobberghen, F., Weiss, H. A., Rahman, A. (2019)	Cluster RCT	Pregnant women (Experiment: n= 283 Control: n= 287)	One-on-one and group counseling to reduce perinatal depression	The depression mean scores (p=0·07) and the prevalence of remission (p=0·14) did not significantly differ between the intervention and control groups at 6 months. However, the intervention group, compared to the control group, reported: (i) reduced depression symptom severity (p<0.0001) and disability score (p<0.0001) at 3 months; (ii) more likely remission (p=0.002) and recovery (p=0.002) at 3 months; and (iii) improvement in perceived social support (p=0.007) at 6 months after childbirth.
Rahman, A., Malik, A., Sikander, S., Roberts, C., & Creed, F. (2008)	RCT	Pregnant women (Experiment: n= 463 Control: n= 440)	Group cognitive behavioral therapy for mothers with depression	The intervention group had lower depression scores at two time points compared to the control group: 1. 6 month point decline (p-value= <0·0001) 2. 12 month point decline (p=<0·0001). Intervention group also showed significant improvement at both time points for (i) overall functioning (6 months= <0.0001; 12 months= <0•0001) and (ii) perception of social support (6 months= <0.0001; 12 months= <0.0001).
Hameed, W., Azmat, S. K., Ali, M., Ishaque, M., Abbas, G., Munroe, E., Ahmed, A. (2016)	Non-inferiority trial without randomisation	Married women using long-acting reversible contraceptive (Experiment: n= 1,246 Control: none)	Compare active (doorstep and telephonic) versus passive (needs-based) follow-up in sustaining use of long-acting reversible contraceptive	Active follow-up is more effective (compared to passive approach) in sustaining long acting reversible contraceptives than the passive (needs-based) follow-up (p= 0.035). Telephone-based follow-up is as effective as the home-based follow-up (compared to passive approach) in sustaining long acting reversible contraceptives (p= 0.431)
Sikander, S., Maselko, J., Zafar, S., Haq, Z., Ahmad, I., Ahmad, M., Rahman, A. (2015)	RCT	Pregnant women (Experiment: n= 210 Control: n= 211)	Cognitive-behavioral counseling to improve rate and duration of exclusive breastfeeding during first 6 months	Increase in exclusive breastfeeding (p = < 0.001) and decline in the use prelacteal feeds (p = < 0.001) in the intervention group over the control group.
Jokhio, A. H., Winter, H. R., & Cheng, K. K. (2005)	Cluster RCT	Pregnant women (Experiment: n= 10,093 Control: n= 9,432)	Training to TBAs to reduce perinatal and maternal mortality and reduce complications of pregnancy	The intervention areas, compared to the control, had lower odds of: (i) perinatal death (p=<0.001) and (ii) maternal mortality of (p=<0.001).

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Omer, K., Mhatre, S., Ansari, N., Laucirica, J., & Andersson, N. (2008)	Cluster RCT	Pregnant women (Experiment: n= 529 Control: n= 541)	Developing community-based communication tools to promote favorable maternal health practices	Women in the intervention communities were more likely to attend prenatal checkups, to stop routine heavy work during pregnancy, to give colostrum to newborn babies, and to maintain exclusive breastfeeding for four months. No p values were reported.
Kumar, R., Farzeen, M., Hafeez, A., Achakzai, B. K., Vankwani, M., Lal, M., Somrongthong, R. (2020)	Non-randomized quasi-experiment	Pregnant women with children up to 6 months of age (Experiment: n= 100 Control: n= 100)	Health education intervention for the prevention of malaria in pregnant women with children up to 6 months of age	The intervention group, compared to the control, showed an increase in: (i) knowledge score (p= < 0.01), and (ii) use of long-lasting insecticide treated bed nets (p < 0.05).
Hirani, S. S., Norris, C. M., Van Vliet, K. J., Van Zanten, S. V., Karmaliani, R., & Lasiuk, G. (2018)	RCT	Young adult women, 20–45 years (Experiment: n= 60 Control: n= 60)	Social support intervention, in groups, to enhance resilience and quality of life	Women in the intervention group reported improvements in: (i) resilience scale-14 item (p = 0.022); (ii) the resilience scale for adults (p = 0.043). No significant findings were noted in the quality of life scores.
Maselko, J., Sikander, S., Turner, E. L., Bates, L. M., Ahmad, I., Atif, N., Rahman, A. (2020)	RCT	Pregnant women (Experiment: n= 206 Control: n= 683)	Group psychosocial intervention to reduce maternal depression	Reduced symptom severity and high remission rates were seen across both the intervention and the control group. Significant outcome differences between the intervention and control group were not found.
Azmat, S. K., Shaikh, B. T., Hameed, W., Mustafa, G., Hussain, W., Asghar, J., Bilgrami, M. (2013)	Non-randomized quasi-experiment	Married women reproductive age (Experiment: n= 2,483 Control: n= 2,509)	Social franchise programme + a free voucher scheme to promote awareness and use of modern long term contraceptive	Improvement in intervention group over the control group in: 1. Awareness of contraception (p < 0.001); 2. Ever use of contraception (p <0.001); 3. Ever use of any modern method (p < 0.0010); 4. Overall contraceptive prevalence rate (p <0.001).
Azmat, S. K., Hameed, W., Hamza, H. B., Mustafa, G., Ishaque, M., Abbas, G., Temmerman, M. (2016)	Non-randomized quasi-experiment	Married women reproductive age (Experiment: n= 1,817 Control: n= 1,075)	Compare two interventions to improve the use of modern contraceptive methods: 1) Social franchise model along with free vouchers and 2) Community Midwife Model	- Both the models were effective for the intervention group over the control group. The Social franchising and free voucher model: (i) contraceptive awareness(p < 0.001), (ii) use of contraceptives (p=<0.0001), (iii) long term modern method-intrauterine device (IUD) use (<0.0001). The Community midwife model: (i) contraceptive awareness (p < 0.001), (ii) use of contraceptives (p=<0.0001) and, (iii) IUD use (0.0078). The social franchising and voucher approach led to a 35 % greater prevalence of contraceptive use in women compared to the CMW model.
Ali, M., Azmat, S. K., Hamza, H. B., & Rahman, M. M. (2020)	Non-randomized quasi-experiment	Married women reproductive age (Experiment: n= 1,276 Control: n= 1,276)	Subsidized multipurpose voucher + family planning counseling to improve contraceptive use, postnatal care and child immunization	There was no increase in modern contraception use in the intervention area (p<0.0001). However, the concentration index and slope index of inequalities for first-time use of modern contraceptives, knowledge of contraceptives, receiving ANC, and delivery at health facilities were negative, indicating that the use of these services was more concentrated among the disadvantaged in intervention areas.
Midhet, F., & Becker, S. (2010)	RCT	Pregnant women (Experiment: n= 1,539 Control: n= 1,022)	Education sessions for safe motherhood practices + TBA training + Emergency transport system to improve maternal and neonatal health indicators	The intervention clusters received prenatal care and prophylactic iron therapy more frequently than pregnant women in control clusters (p=<0.05). Providing safe motherhood education resulted in significant increase in percent of hospital deliveries (p=<0.05) and reduction in perinatal mortality (p=<0.05). There was no impact on the use of skilled birth attendants.
Qureshi, R. N., Sheikh, S., Hoodbhoy, Z., Sharma, S., Vidler, M., Payne, B. A., Bhutta, Z. A. (2020)	RCT	Pregnant women screened for preeclampsia (Experiment: n= 1,276 Control: n= 1,276)	Mobile health assessment+ referral+ doorstep + educational sessions to reduce all-cause maternal & perinatal mortality & major morbidity	There was no difference between the groups in the primary outcome of composite maternal, fetal and newborn mortality and major morbidity (p= 0·31). However, there was a reduction in stillbirths (p = 0·03) and no adverse events were reported in the intervention group.

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Effectiveness of primary healthcare interventions

Due to the heterogeneity in terms of the included study designs and interventions, a narrative report of effectiveness has been provided, organized according to the type of intervention.

Counseling Interventions

Eight included studies evaluated the effectiveness of counseling interventions. Six of these interventions targeted improvement in mental health; and two aimed to improve family planning or breastfeeding practices.

i) Mental health

Only one of the six mental health interventions aimed to deliver counseling to all women of any age who had screened positive for anxiety and/or depression. This RCT was conducted with women of a semi-urban community in Karachi city and the intervention consisted of one-to-one counseling sessions delivered by briefly trained women from the community 27. The women in the intervention group received eight sessions of counseling, which included supportive, problem-solving, and cognitive-behavioral techniques. The results showed a significant reduction in both the experimental and control groups post intervention, this possibly having been due to some women in the control group also receiving counseling.

Two studies involved the delivery of services to women affected by conflict in the Swat district, and who screened positive for mental health disorders 28 32. Both interventions were group-based programs, involving adaptions of the World Health Organization Problem Management intervention (PM+), and delivered by briefly trained female non-specialists and LHWs. Both cluster RCTs reported a significant decline in anxiety and depression post intervention.

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One RCT involved the deliver of one-to-one counseling sessions to mothers with children between 0-30 months, screened for postnatal depression 21. This non-randomized quasi-experimental study included women from two underdeveloped colonies of Karachi city and the one-to-one counseling was delivered by briefly trained women from the community. The intervention consisted of supportive, problem-solving and basic cognitive behavioral techniques delivered for eight weeks in weekly one-hour sessions. The study found a significant decline in anxiety and depression in both the counseled and the non-counseled groups. This may be because the control group received enhanced social support through regular visits by the counselors although the number of visits was not described. In addition, the study reported a greater decline in the counseled group, compared to the non-counseled for: (i) recovery; (ii) reduction in the rate of recurrence; and (iii) increase in the duration before relapse.

Two mental health intervention studies evaluated the effectiveness of cognitive behaviour therapy (CBT) in supporting women experiencing maternal depression. In a cluster RCT, involving 40 villages from a rural district of Rawalpindi city, pregnant women who screened for positive for depression were recruited to the trial 30. The Thinking Healthy Programme (THP) for perinatal depression was used, which includes a combination of strategies that incorporate behavioural activation, active listening, collaboration with the family, guided discovery, and homework, and was delivered by briefly trained volunteer peers from the community with LHWs as facilitators. The intervention consisted of ten individual and four group sessions, from the third trimester of pregnancy to 6 months after childbirth. The intervention group, compared to the control group, reported significant improvements in: (i) depression symptom severity and disability score at 3 months; (ii) remission and recovery at 3 months; and (iii) perceived social support at 6 months after childbirth.

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The second RCT involved the delivery of CBT to women from 40 villages from the rural district of Rawalpindi city 15. Pregnant women screened for depression received the THP intervention, in group sessions, delivered by specially trained LHWs. Significantly lower depression scores were reported for the intervention group, at the 6- and 12-month time points. Additionally, mothers in the intervention group showed significant improvement in overall functioning and perception of social support, at both 6 and 12 month time points. Both studies were found to be cost-effective for resource-poor nations as they utilized volunteer peers to deliver the intervention.

ii) Family planning practices

Two included studies used counseling to improve family planning practices. In one non-inferiority trial without randomisation, a comparison was made between active and passive follow-up of existing long-acting reversible contraceptive-use (LARC) 22. Married women using long-acting reversible contraception were recruited from twenty-two rural and urban primary healthcare facilities of Chakwal city. Female health educators delivered one-to-one sessions using an adapted manual for family planning services based on the Family Advancement for Life and Health Project. Results showed that both the telephone-based follow up and home-based visits significantly improved the use of LARC, compared to the passive or needs-based follow-up.

The second study, an RCT, evaluated the effectiveness of an intervention to improve temporary contraceptive methods through breast-feeding practices. The intervention was delivered by LHWs using one-to-one cognitive-behavioral counseling sessions aimed at improving the rate and duration of exclusive breastfeeding during the first 6 months of an infant's life 31. The results showed a significant increase in the duration of exclusive breast-

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feeding and also in rates of exclusive breast-feeding at six months postpartum in the intervention group. The study also reported a decline in the use of prelacteal feeds to infants in the intervention group.

Health Education and Awareness Interventions

Three studies evaluated the effectiveness of health education and awareness interventions. Two of these involved the use of cluster RCTs to improve maternal health indicators and practices. One cluster RCT involved training traditional birth attendants (TBAs), and also providing them with safe delivery kits, across seven sub-districts in Larkana city 34. A team of obstetricians and female paramedics trained the TBAs; and LHWs were further trained to provide support to the TBAs in the community. The training involved the use of picture cards about safe practices for antepartum, intrapartum, and postpartum care. Additionally, they were trained in how to conduct a clean delivery; use of the disposable delivery kit; and when to refer women for emergency obstetrical care. The results showed significantly lower odds of perinatal death and maternal mortality in the intervention group.

A second cluster RCT involved the delivery of educational sessions to pregnant women or mothers with young children about safe maternal practices 35. Women from ten underdeveloped areas from three cities of Sindh province were included and the awareness communication tool was developed in consultation sessions between LHWs and government health officials. It was decided that embroidered pictures on a traditional Sindh fabric (known as *ajrak*) should be used to communicate user-friendly, indigenous, and attractive messages to raise awareness about the importance of antenatal check-ups, giving colostrum after birth, and not doing heavy work during pregnancy. The intervention group showed an increase in prenatal

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checkups, stopping routine heavy work during pregnancy, provision of colostrum to newborns, and exclusive breastfeeding.

The third study evaluated the effectiveness of a health education and awareness intervention aimed at improving malaria net use in pregnant women 23. This non-randomized quasi-experimental study included women from two areas of Tarparkar district. LHWs were responsible for providing women with information about malaria prevention and the importance of using long-lasting in-bed nets (LLIN). Results showed a significant increase in the experimental group's knowledge and use of LLIN.

Social and Psychosocial Interventions

Two RCTs involved an evaluation of the effectiveness of interventions providing social or psychosocial support to improve resilience and quality of life, and to reduce maternal depression, respectively. The first study was delivered to young adult women, 20-45 years, in low socioeconomic areas of Karachi city 36. Women were provided with social support in six weekly group sessions, using a manual that guided women about stress management and coping strategies for stress. Results revealed a statically significant improvement in resilience for the intervention group, but no impact on the quality of life (QOL) indicators.

The second study, assessed the effectiveness of a psychosocial intervention in pregnant women screened for depression 38. The intervention was delivered across forty village clusters of Rawalpindi city by non-specialist volunteer peers who were lay married women living in the same community. Both the intervention and control groups received enhanced usual care from the LHW program, which consisted of: (i) informing participants about their depression status and ways to seek help for it and also providing them with an information leaflet, (ii) informing

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LHWs about each woman's depression status, and (iii) training the primary care facility-based physicians in the sub-district in the management of maternal depression. In addition, the intervention group received a longer-duration, psychosocial intervention, consisting of 18 group-based sessions (from 7 months to 36 months postnatal) that included peer-support, behavioral activation, and problem solving in a culturally sensitive and non-medicalised format. The findings showed a decline in maternal depression symptoms and remission in both the intervention and enhanced care groups, thereby masking any potential benefits of the intervention.

Combination Interventions

Five studies evaluated the effectiveness of combination interventions on a) family planning and contraception use; or b) wider maternal health indicators. The combined interventions included: (i) Voucher system *plus* educational outreach, (ii) Education sessions for safe motherhood practices, *plus* TBA training, *plus* emergency transport system; and (iii) Mobile health-guided clinical assessment, *plus* referral to facility for specialized treatment, *plus* educational sessions.

i) Family planning and contraception use

Three studies evaluated the effectiveness of family planning and contraceptive use. The first of these was a quasi-experimental study, which included married women of reproductive age from rural and under-served communities of two provinces - Punjab and Sindh [24]. A social franchise programme delivered by female health visitors, midwives and nurses, along with a free voucher scheme, was used to promote awareness and use of modern long-term contraceptives. Results showed a statistically significant improvement in the intervention group for (i) awareness of

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modern contraception, (ii) ever use of modern contraceptives, (iii) overall contraceptive prevalence rate, and (iv) uptake of intrauterine devices.

A second quasi-experimental study compared a social franchising and voucher model with educational sessions provided by CMWs 25. The study included married women of reproductive age from eight districts of three provinces - Sindh, Punjab and KPK. Trained facilitators and CMWs delivered the intervention and results showed a significant increase in contraception awareness, contraception use, and long term modern method-intrauterine device through both the social franchising and voucher approach and the CMW model. However, the social franchising and voucher approach led to a 35% greater uptake of contraceptive use.

A third non-randomized quasi-experimental study assessed the effectiveness of a subsidized multipurpose voucher *plus* family planning counseling to improve contraceptive use and postnatal care 43. Married women of reproductive age were included from one city each from urban and rural Punjab. The intervention was delivered by community service providers who were trained to provide family planning counseling and supported by local distributing agencies who provided vouchers. Although there was no net increase in modern contraception use there was a low discontinuation rate, and vaccination rates for BCG increased significantly by 14%, and 5% each for DPT, HBV, and measles. Furthermore, use of services was more concentrated among the disadvantaged in intervention areas than the wealthy counterparts, thereby suggesting increased equity.

ii) Maternal health indictors

Two RCTs evaluated the effectiveness of multicomponent interventions on maternal health indicators. The first RCT evaluated the effectiveness of education sessions promoting safe

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motherhood practices *plus* TBA training *plus* an emergency transport system 41. Both pregnant women and their husbands received the intervention across thirty-two village clusters of Khuzdar city, through the combined efforts of local women and TBAs. Results showed a significant increase in hospital deliveries and a reduction in perinatal mortality in the intervention groups. Additionally, the intervention clusters received prenatal care and prophylactic iron therapy more frequently than pregnant women in the control clusters.

The second RCT involved an evaluation of the effectiveness of mobile health-guided clinical assessment *plus* referral to a facility for specialized treatment *plus* educational sessions 42. Pregnant women screened for preeclampsia were recruited from two cities of the Sindh province. LHWs delivered the intervention and received support from nurses and doctors at primary health centers as and when referrals were made based on need. While the results showed no impact on the composite measure of maternal health indicators, other benefits of the intervention were found, such as a statistically significant reduction in stillbirths, and no adverse events were recorded for the intervention group.

DISCUSSION

This is the first systematic review to synthesise evaluations of primary care interventions that are aimed at improving women's health outcomes. Such reviews are essential to the development of effective healthcare policy especially in developing regions 44. Furthermore, country-specific systematic reviews of this nature are best able to identify regional effectiveness that is central to guiding the developing of such health policy 45. Four types of intervention were identified by this review- Counseling; Health Education and Awareness; Social and Psychosocial; and Combination Interventions (i.e. vouchers, education, referral and technology) – and all included studies reported favorable or partially favorable results (i.e. on secondary outcomes) pointing to

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the value of providing such care in terms of improving the health of women in Pakistan. However, most of the included studies evaluated the effectiveness of interventions to women of reproductive years and only four studies evaluated interventions that were not targeting reproductive health or family planning. These findings are consistent with the wider literature, which indicates that interventions for maternal health are given comparative priority, as opposed to health services for other groups such as unmarried women, aging women, and those with special needs or disabilities 46. In addition, a large number of health conditions appear not to be addressed as part of primary level interventions including (i) chronic disease and multimorbidity 47, and (ii) violence and injury 48. While the focus of primary healthcare services on women's reproductive health in developing countries and conservative climates may reflect the high maternal and child mortality that is present 49, the failure to address the wider health care needs of Pakistan's population of women, leaves a significant health burden both for the women themselves and society more widely.

With regard to the effectiveness of counseling interventions, nine studies showed evidence of effectiveness in terms of their primary goal of improving mental health, and family planning or maternal health practices. Both one-to-one and group-based counseling sessions were successful in improving women's health outcomes, and the results also showed that an active home or telephone-based follow-up was more effective compared to a passive needs-based approach 22, in terms of reinforcing and sustaining behavioural change 50, which is important in a context where a majority of the population are disadvantaged and semi-literate.

The studies of counseling interventions also demonstrated the effectiveness of using low-cost and more acceptable community peers as intervention facilitators, both important points for a culturally conservative and resource poor country like Pakistan 51. Furthermore, there were

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also benefits of awareness building and social support as part of routine visits, which was especially valuable for disadvantaged and semi-literate women who needed reaffirmation and regular care contact 52. It is noteworthy that two studies in this review showed improvement in the mental health of Pakistani women impacted by conflict. Given that a large proportion of people in Pakistan are affected by conflict and natural disasters 53, these findings suggest that such interventions should be integrated into routine primary care services.

The three health education and awareness interventions reported effectiveness in improving preventive practices for Malaria, training of existing TBAs, and the development of community-based tools to advance maternal health practices. Training of TBAs and development of culturally acceptable tools is important in terms of the acceptance and uptake of health services for women in a conservative region like Pakistan. In Pakistani society both culture and tradition compel women to seek health services from TBAs as the first line of support 54, and as such, the training of TBAs provides an important way of improving the reproductive of health of Pakistani women.

The psychosocial interventions were found to improve resilience and quality of life for women, which is important in a patriarchal culture where women face barriers to mobility and access to social support 55. Preventive health care of this nature is critical in terms of improving the wellbeing of women and reducing the national health burden 56. The group psychosocial intervention served as an effective way of utilizing peer-support and problem-solving in a culturally sensitive and non-medicalised format. Wider research confirms that both group sessions and problem-solving are valuable tools to improve health behaviors and preventive practices in women 57.

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All five combination interventions were focused on reproductive health and family planning. The findings from this group of interventions suggest that educational counseling and door-to-door services need to be coupled with financial interventions, such as free or subsidized voucher schemes. Despite the provision of free community services, optimal healthcare outcomes are only possible when financial support is available for transport, private consultancy, and purchase of medicines or other health commodities [58]. For low income nations such as Pakistan, free or subsidized vouchers are integral to effective uptake of services and better health outcomes [25 39]. Though education, awareness and counseling are effective interventions, many poor women are prevented from using them without financial subsidies [24 43] and the provision of transport services [41].

One combination intervention highlighted the effectiveness of delivering awareness and training to husbands, which is important to enable women in the region to access services, travel for services, and return for continued care. The patriarchal climate of the region, with men as primary decision-makers for women's health behavior and mobility, makes interventions with husbands more effective for women's health outcomes [59].

Overall, the results support the use of (i) an active approach, (ii) delivery by community peers, and (iii) financial vouchers. A number of studies showed that an active approach involving door-to-door visitation [22 40], is also helpful for building general health awareness and social support [21 36] and improving preventive health [23 33]. The training of local peers also appears to be a cost-effective and highly acceptable way to support women living in conservative communities [15 27 30 34 41]. Furthermore, the evidence suggests that locally trained community women can be even more successful in improving health outcomes when they are partnered with existing government deployed LHWs [28 32]. However, employing briefly trained non-specialists

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from the community requires fidelity measures and monitoring to ensure effective and consistent delivery of services [29] [38].

Limitations

The diverse nature of the interventions included in this review, alongside the use of different outcomes makes a comparison of the included interventions difficult. Furthermore, we included seven studies that were not RCTs, and as such cannot be confident that the results of these studies do not reflect a regression to the mean. Overall, the quality of the included studies was good to moderate. The assessment of outcome for all included studies relied on the use of patient-reported outcomes, which is of concern especially in underdeveloped regions with illiterate and semi-literate populations who may not answer the survey or self-report correctly, even with the assistance of data collectors, due to low levels of understanding [60]. It is now recognised that health intervention assessments must combine patient-reported outcomes with clinical outcomes in order to provide more robust evidence about health outcomes in women [61].

Most of the studies included in this review were conducted in either one city or one province, and as a result of the fact that the cultural climate and behavioral response for each province of Pakistan is unique [62], it is not possible to make generalizations about the effectiveness of interventions across Pakistan based on the current evidence. Only when interventions are delivered across provinces and remote locations will it be possible to generalize the findings. This will also ensure that the designing of services and policies are ethnically and regionally appropriate and based on the different needs of diverse population groups [63]. As discussed earlier, the studies focused largely on women of reproductive age or pregnant women and the interventions are limited to the delivery of services for family planning and maternal health.

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Conclusion

Efforts to enhance the nature and quality of women's primary healthcare services in Pakistan must be informed by contextualized evidence. This systematic review found that to date, evidence regarding the effectiveness of primary care services for women in Pakistan remains limited and focused primarily on reproductive or mental health in only one or two provinces. There is, as such, an absence of information about the effectiveness of primary care services for older women, women with special needs, refugees and displaced women, and unmarried women. This review has highlighted the effectiveness of a number of primary care based services and the benefit of active door-to-door and group-based approaches; utilization of community peers who can deliver care cost-effectively and who are more accepted in the community; and the benefit of integrating financial vouchers to support uptake in poor populations.

Contribution Statement

SRJ and JB conceptualized the study, designed the methodology, and wrote the manuscript. SRJ, QKM, SM, and MA performed the literature search. SRJ interpreted the data and JB contributed to the interpretation. JB supervised the project.

Competing Interests

There are no competing interests to declare.

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Data sharing statement

Data sharing not applicable as no datasets generated and/or analysed for this study. All data relevant to the study are included in the article or uploaded as online supplemental information.

Patient and public involvement

No patient involved

Ethics approval

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This study does not involve human participants. It is a systematic review of studies and does not require ethics approval from an IRB.

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Supplementary Files:
Supplementary File 1: Search Terms and Database Results
Supplementary File 2: Quality Appraisal of Included Studies
Supplementary File 3: Characteristics of Included Studies

Figure Legend

Figure 1- Preferred Reporting Items for Systematic Reviews and Meta-Analyses flow chart.

REFERENCES

1. Nishtar S, Boerma T, Amjad S, et al. Pakistan's health system: performance and prospects after the 18th Constitutional Amendment. *J The Lancet* 2013;381(9884):2193-206.
2. World Bank. Pakistan Development Update. 2017
3. United Nations Pakistan. One United Nations Programme III 2018-2022, United Nations Sustainable Development Framework for Pakistan., 2018.
4. Mumtaz Z, Salway S, Waseem M, et al. Gender-based barriers to primary health care provision in Pakistan: the experience of female providers. *J Health policy planning* 2003;18(3):261-69.
5. Aziz A, Saleem S, Nolen TL, et al. Why are the Pakistani maternal, fetal and newborn outcomes so poor compared to other low and middle-income countries? *J Reproductive Health* 2020;17(3):1-12.
6. Jafree SR. Determinants of depression in women with chronic disease: Evidence from a sample of poor loan takers from Pakistan. *J Journal of Community Psychology* 2020;48(7):2238-51.
7. Butt M, Mohammed R, Butt E, et al. Why have immunization efforts in Pakistan failed to achieve global standards of vaccination uptake and infectious disease control? *J Risk management healthcare policy* 2020;13:111.
8. Ali PA, Gavino MIB. Violence against women in Pakistan: A framework for analysis. *J Journal-Pakistan Medical Association* 2008;58(4):198.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

9. Khan MA, Jafree SR, Jibeen T. Social Barriers to Mental Well-Being in Women of South Asia. *J The Sociology of South Asian Women's Health* 2020:129-50.
10. Jalal S, Younis MZ. Aging and elderly in Pakistan. *J Ageing International* 2014;39(1):4-12.
11. NIPS. Health Survey (PDHS)(2012-13)- National Institute of Population Studies,. 2017
12. Health Planning SSaIAUH, Ministry of National Health Services RaC. Pakistan Monitoring Report - Universal Health Coverage, 2021.
13. Jalal S. The lady health worker program in Pakistan—a commentary. *J European journal of public health* 2011;21(2):143-44.
14. Douthwaite M, Ward P. Increasing contraceptive use in rural Pakistan: an evaluation of the Lady Health Worker Programme. *J Health policy planning* 2005;20(2):117-23.
15. Rahman A, Malik A, Sikander S, et al. Cognitive behaviour therapy-based intervention by community health workers for mothers with depression and their infants in rural Pakistan: a cluster-randomised controlled trial. *J The Lancet* 2008;372(9642):902-09.
16. Bhutta ZA, Memon ZA, Soofi S, et al. Implementing community-based perinatal care: results from a pilot study in rural Pakistan. *J Bulletin of the World Health Organization* 2008;86:452-59.
17. Shaikh BT, Hatcher J. Health seeking behaviour and health service utilization in Pakistan: challenging the policy makers. *J Journal of public health* 2005;27(1):49-54.
18. Hasnain S, Sheikh N. Causes of low tetanus toxoid vaccination coverage in pregnant women in Lahore district, Pakistan. *%J EMHJ-Eastern Mediterranean Health Journal, 13* 2007
19. Jafree SR, Mahmood QK, ul Momina A, et al. Protocol for a systematic review of barriers, facilitators and outcomes in primary healthcare services for women in Pakistan. *J BMJ open* 2021;11(3):e043715.
20. University of Oxford. Randomised Controlled Trials (RCT) Critical Appraisal Sheet. 2021
21. Ali NS, Ali BS, Azam IS, et al. Effectiveness of counseling for anxiety and depression in mothers of children ages 0-30 months by community workers in Karachi, Pakistan: a quasi experimental study. *J BMC psychiatry* 2010;10(1):1-9.
22. Hameed W, Azmat SK, Ali M, et al. Comparing effectiveness of active and passive client follow-up approaches in sustaining the continued use of long acting reversible contraceptives (LARC) in Rural Punjab: a multicentre, non-inferiority trial. *J PloS one* 2016;11(9):e0160683.
23. Kumar R, Farzeen M, Hafeez A, et al. Effectiveness of a health education intervention on the use of long-lasting insecticidal nets for the prevention of malaria in pregnant women of Pakistan: a quasi-experimental study. *J Malaria Journal* 2020;19(1):1-10.
24. Azmat SK, Shaikh BT, Hameed W, et al. Impact of social franchising on contraceptive use when complemented by vouchers: a quasi-experimental study in rural Pakistan. *J PLoS One* 2013;8(9):e74260.
25. Azmat SK, Hameed W, Hamza HB, et al. Engaging with community-based public and private mid-level providers for promoting the use of modern contraceptive methods in rural Pakistan: results from two innovative birth spacing interventions. *J Reproductive health* 2016;13(1):1-15.
26. Ariff S, Khan U, Turab A, et al. Health care seeking behaviors in Puerperal sepsis in rural Sindh, Pakistan: a qualitative study. *J medRxiv* 2020
27. Ali BS, Rahbar MH, Naeem S, et al. The effectiveness of counseling on anxiety and depression by minimally trained counselors: a randomized controlled trial. *J American Journal of Psychotherapy* 2003;57(3):324.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

28. Rahman A, Khan MN, Hamdani SU, et al. Effectiveness of a brief group psychological intervention for women in a post-conflict setting in Pakistan: a single-blind, cluster, randomised controlled trial. *J The Lancet* 2019;393(10182):1733-44.
29. Hackett K, Henry E, Hussain I, et al. Impact of home-based family planning counselling and referral on modern contraceptive use in Karachi, Pakistan: a retrospective, cross-sectional matched control study. *J BMJ open* 2020;10(9):e039835.
30. Sikander S, Ahmad I, Atif N, et al. Delivering the Thinking Healthy Programme for perinatal depression through volunteer peers: a cluster randomised controlled trial in Pakistan. *J The Lancet Psychiatry* 2019;6(2):128-39.
31. Sikander S, Maselko J, Zafar S, et al. Cognitive-behavioral counseling for exclusive breastfeeding in rural pediatrics: a cluster RCT. *J Pediatrics* 2015;135(2):e424-e31.
32. Khan M, Hamdani S, Chiumento A, et al. Evaluating feasibility and acceptability of a group WHO trans-diagnostic intervention for women with common mental disorders in rural Pakistan: a cluster randomised controlled feasibility trial. *J Epidemiology psychiatric sciences* 2017;28(1):77-87.
33. Shallwani K, Ramji R, Ali TS, et al. Self-examination for breast and testicular cancers: a community-based intervention study. *J Asian Pacific journal of cancer prevention: APJCP* 2010;11(1):145.
34. Jokhio AH, Winter HR, Cheng KK. An intervention involving traditional birth attendants and perinatal and maternal mortality in Pakistan. *J New England Journal of Medicine* 2005;352(20):2091-99.
35. Omer K, Mhatre S, Ansari N, et al. Evidence-based training of frontline health workers for door-to-door health promotion: a pilot randomized controlled cluster trial with Lady Health Workers in Sindh Province, Pakistan. *J Patient education counseling* 2008;72(2):178-85.
36. Hirani SS, Norris CM, Van Vliet KJ, et al. Social support intervention to promote resilience and quality of life in women living in Karachi, Pakistan: a randomized controlled trial. *J International journal of public health* 2018;63(6):693-702.
37. Khushk WA, Fatmi Z, White F, et al. Health and social impacts of improved stoves on rural women: a pilot intervention in Sindh, Pakistan. *J Indoor air* 2005;15(5):311-16.
38. Maselko J, Sikander S, Turner EL, et al. Effectiveness of a peer-delivered, psychosocial intervention on maternal depression and child development at 3 years postnatal: a cluster randomised trial in Pakistan. *J The Lancet Psychiatry* 2020;7(9):775-87.
39. Agha S. Impact of a maternal health voucher scheme on institutional delivery among low income women in Pakistan. *J Reproductive health* 2011;8(1):1-14.
40. Najmi H, Ahmed H, Halepota G, et al. Community-based integrated approach to changing women's family planning behaviour in Pakistan, 2014–2016. *J Public health action* 2018;8(2):85-90.
41. Midhet F, Becker S. Impact of community-based interventions on maternal and neonatal health indicators: Results from a community randomized trial in rural Balochistan, Pakistan. *J Reproductive health* 2010;7(1):1-10.
42. Qureshi RN, Sheikh S, Hoodbhoy Z, et al. Community-Level Interventions for Pre-eclampsia (CLIP) in Pakistan: a cluster randomised controlled trial. *J Pregnancy hypertension* 2020;22:109-18.
43. Ali M, Azmat SK, Hamza HB, et al. Assessing Effectiveness of Multipurpose Voucher Scheme to Enhance Contraceptive Choices, Equity, and Child Immunization Coverage:

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Results of an Interventional Study from Pakistan. *J Journal of Multidisciplinary Healthcare* 2020;13:1061.

44. Kringos DS, Boerma WG, Hutchinson A, et al. The breadth of primary care: a systematic literature review of its core dimensions. *J BMC health services research* 2010;10(1):1-13.
45. Politzer R, Regan J, Shi L, et al. Primary care quality: community health center and health maintenance organization. *J Southern medical journal* 2003;96(8):787-95.
46. Mirza I, Tareen A, Davidson L, et al. Community management of intellectual disabilities in Pakistan: a mixed methods study. *J Journal of Intellectual Disability Research* 2009;53(6):559-70.
47. Singh K, Patel SA, Biswas S, et al. Multimorbidity in South Asian adults: prevalence, risk factors and mortality. *J Journal of Public Health* 2019;41(1):80-89.
48. Zakar R, Zakar, M. Z., & Kraemer, A. . Primary health care physicians' response to the victims of spousal violence against women in Pakistan. . *Health care for women international, 32(9), 811-832* 2011
49. Habumuremyi P D ZM. "Making Family Planning a National Development Priority." *The Lancet 380 (9837): 78–80* 2012
50. Bruce J. Fundamental elements of the quality of care: a simple framework. *J Studies in family planning* 1990;21(2):61-91.
51. Sheikh KS, Ahmad K, Farooq A. Socio-Political and Cultural Challenges of Poverty Alleviation: A Comparative Analysis of Pakistan and Bangladesh. *J South Asian Studies* 2016;31(2):291.
52. Malik MN, Awan MS, Saleem T. Social mobilization campaign to tackle immunization hesitancy in Sargodha and Khushab districts of Pakistan. *J Journal of Global Health* 2020;10(2)
53. Naqvi ZZ, Riaz S. Women in Pakistan: Countering conflicts and building peace. *J Asian Journal of Women's Studies* 2015;21(3):326-38.
54. Omer S, Zakar R, Zakar MZ, et al. The influence of social and cultural practices on maternal mortality: a qualitative study from South Punjab, Pakistan. *J Reproductive health* 2021;18(1):1-12.
55. Mumtaz Z, Salway S. 'I never go anywhere': extricating the links between women's mobility and uptake of reproductive health services in Pakistan. *J Social science medicine* 2005;60(8):1751-65.
56. Zaidi S, Saligram P, Ahmed S, et al. Expanding access to healthcare in South Asia. *J BMJ open* 2017;357
57. D'Zurilla TJ, Nezu AM. Problem-solving therapy. *J Handbook of cognitive-behavioral therapies* 2010;3:197-225.
58. Di Cesare M, Bhatti Z, Soofi SB, et al. Geographical and socioeconomic inequalities in women and children's nutritional status in Pakistan in 2011: an analysis of data from a nationally representative survey. *J The Lancet Global Health* 2015;3(4):e229-e39.
59. Ahmed S, Jafri H, Rashid Y, et al. Autonomous decision-making for antenatal screening in Pakistan: views held by women, men and health professionals in a low–middle income country. *J European Journal of Human Genetics* 2019;27(6):848-56.
60. Jahagirdar D, Kroll T, Ritchie K, et al. Using patient reported outcome measures in health services: a qualitative study on including people with low literacy skills and learning disabilities. *J BMC health services research* 2012;12(1):1-8.

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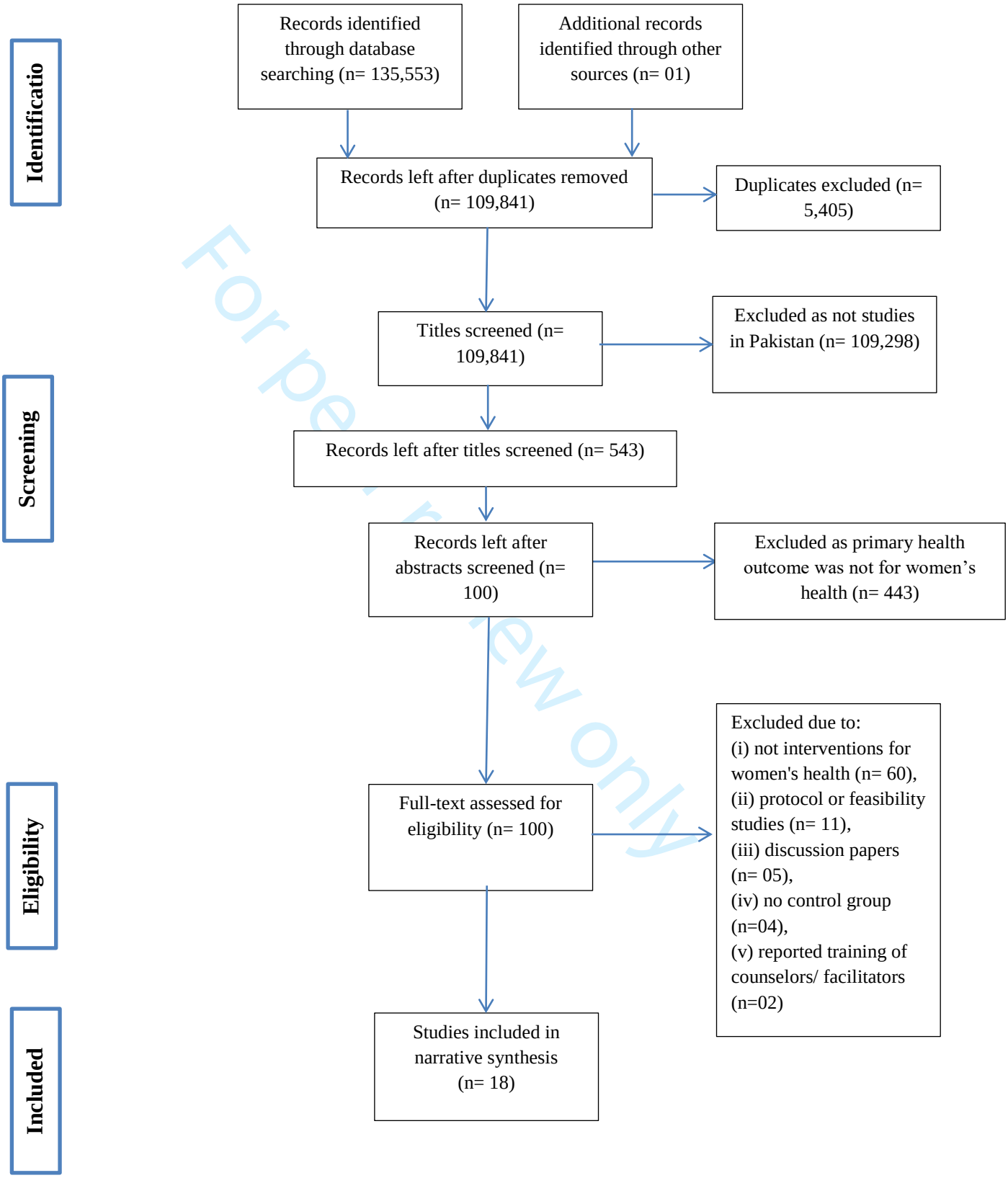
61. Meadows KA. Patient-reported outcome measures: an overview. *J British journal of community nursing* 2011;16(3):146-51.
62. Muhammad Gadit AA, Mugford G. Prevalence of depression among households in three capital cities of Pakistan: need to revise the mental health policy. *J Plos one* 2007;2(2):e209.
63. Londoño J-L, Frenk J. Structured pluralism: towards an innovative model for health system reform in Latin America. *J Health Policy* 1997;41(1):1-36.

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Figure 1-

Preferred Reporting Items for Systematic Reviews and Meta-Analyses flow chart



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Supplementary File 1-

Search Terms and Database Results

Search terms

1. primary health OR community health AND Pakistan AND women

2. primary healthcare OR primary health care OR primary service delivery OR primary health service* OR primary health evaluation OR program* OR polic* AND Pakistan AND women

3. primary health OR community health OR wom#n OR mother OR maternal OR women's health OR maternal health OR reproductive health AND Pakistan AND women

4. primary health OR community health OR evidence based practice OR intervention OR RCTs OR randomi*ed controlled trial* OR cluster randomi*ed controlled trial* OR clinical trial* OR quasi-experimental OR non-randomi*ed OR cross-sectional OR mixed methods OR quantitative OR qualitative OR case-study OR group studies OR evidence based OR cohort studies OR case-control studies AND Pakistan AND women

5. primary health OR community health OR (lady health worker OR LHW) OR (community health worker OR CHW) OR (community midwife OR CMW) OR (traditional birth attendant OR TBA) AND Pakistan AND women

Table:
Summary of results from databases searched, as at August ….

Search Term	PubMed	BMC	Medline	CINAHL	Cochrane Library	Total
1.	4,074	2,435	207	1,756	1,679	10,151
2.	1,702	1,769	8,088	2,522	2,438	16,519
3.	4,427	2,435	207	2,686	2,597	12,352
4.	5,029	2,436	23,894	7,819	7,688	46,866
5.	5,458	595	52	0	8,340	14,445
6.	5,340	2,435	23,665	1,931	1,849	35,220
	26,030	**12,105**	**56,113**	**16,714**	**24,591**	**135,553**

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Supplementary File 2:

Quality Appraisal of Included Studies

The RCT Critical Appraisal Checklist was used.

Key to table 1 screening questions:

1a. R- Was the assignment of patients to treatments randomized? (Yes, No, Unclear)

1b. R- Were the groups similar at the start of the trial? (Yes, No, Unclear)

2a. A – Aside from the allocated treatment, were groups treated equally? (Yes, No, Unclear)

2b. A – Were all patients who entered the trial accounted for? And were they analysed in the groups to which they were randomised? (Yes, No, Unclear)

3. M - Were measures objective or were the patients and clinicians kept "blind" to which treatment was being received? (Yes, No, Unclear)

4. Are there suitable measures used to report results and are the estimates of the treatment effect precise? (Yes, No, Unclear)

Table 1:

					SCREENING QUESTIONS						Score	
No.	**First author**	**Year**	**Type of Study**	**Citation**	**1a. R**	**1b. R**	**2a. A**	**2b. A**	**3. M**	**4**	**High/ Medium/ Low Quality**	**COMMENTS**
1	Ali, B. S	2003	RCT	Ali, B. S., Rahbar, M. H., Naeem, S., Gul, A., Mubeen, S., & Iqbal, A. (2003). The effectiveness of counseling on anxiety and depression by minimally trained counselors: a randomized controlled trial. Am J Psychother, 57(3), 324-336. doi:10.1176/appi.psychotherapy.2003.57.3.324	Yes	Yes	Yes	Yes	Yes	Yes	High	
2	Khan, M. N.	2017	Cluster RCT	Khan, M. N., Hamdani, S. U., Chiumento, A., Dawson, K., Bryant, R. A., Sijbrandij, M., Rahman, A. (2017). Evaluating feasibility and acceptability of a group WHO trans-diagnostic intervention for women with common mental disorders in rural Pakistan: a cluster randomised controlled feasibility trial. Epidemiol Psychiatr Sci, 28(1), 77-87. doi:10.1017/s2045796017000336	Yes	Yes	Yes	Yes	Yes	Yes	High	

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3	Rahman, A	2019	Cluster RCT	Rahman, A., Khan, M. N., Hamdani, S. U., Chiumento, A., Akhtar, P., Nazir, H., van Ommeren, M. (2019). Effectiveness of a brief group psychological intervention for women in a post-conflict setting in Pakistan: a single-blind, cluster, randomised controlled trial. Lancet, 393(10182), 1733-1744. doi:10.1016/s0140-6736(18)32343-2	Yes	Yes	Yes	Yes	Yes	Yes	High	
4	Ali, N. S	2010	Non-randomized quasi-experiment	Ali, N. S., Ali, B. S., Azam, I. S., & Khuwaja, A. K. (2010). Effectiveness of counseling for anxiety and depression in mothers of children ages 0-30 months by community workers in Karachi, Pakistan: a quasi-experimental study. BMC Psychiatry, 10, 57. doi:10.1186/1471-244x-10-57	No	-	-	-	-	Yes	Medium	This was a non-randomized sample from an action research program.
5	Sikander, S.	2019	Cluster RCT	Sikander, S., Ahmad, I., Atif, N., Zaidi, A., Vanobberghen, F., Weiss, H. A., Rahman, A. (2019). Delivering the Thinking Healthy Programme for perinatal depression through volunteer peers: a cluster randomised controlled trial in Pakistan. Lancet Psychiatry, 6(2), 128-139. doi:10.1016/s2215-0366(18)30467-x	Yes	Yes	Yes	Yes	Yes	Yes	High	
6	Rahman, A.	2008	RCT	Rahman, A., Malik, A., Sikander, S., Roberts, C., & Creed, F. (2008). Cognitive behaviour therapy-based intervention by community health workers for mothers with depression and their infants in rural Pakistan: a cluster-randomised controlled trial. Lancet, 372(9642), 902-909. doi:10.1016/s0140-6736(08)61400-2	Yes	Yes	Yes	Yes	Yes	Yes	High	
7	Hameed, W	2016	Non-inferiority trial without randomisation	Hameed, W., Azmat, S. K., Ali, M., Ishaque, M., Abbas, G., Munroe, E., Ahmed, A. (2016). Comparing Effectiveness of Active and Passive Client Follow-Up Approaches in Sustaining the Continued Use of Long Acting Reversible Contraceptives (LARC) in Rural Punjab: A Multicentre, Non-Inferiority Trial. PLoS One, 11(9), e0160683. doi:10.1371/journal.pone.0160683	No	-	-	-	-	Yes	Medium	Union council areas were sampled randomly. Participants could not be randomized, (due to the low proportion of LARC users with access to phones)
8	Sikander, S.	2015	RCT	Sikander, S., Maselko, J., Zafar, S., Haq, Z., Ahmad, I., Ahmad, M., Rahman, A. (2015). Cognitive-behavioral counseling for exclusive breastfeeding in rural pediatrics: a cluster RCT. Pediatrics, 135(2), e424-431. doi:10.1542/peds.2014-1628	Yes	Yes	Yes	Yes	Yes	Yes	High	
9	Jokhio, A. H	2005	Cluster RCT	Jokhio, A. H., Winter, H. R., & Cheng, K. K. (2005). An intervention involving traditional birth attendants and perinatal and maternal mortality in Pakistan. N Engl J Med, 352(20), 2091-2099. doi:10.1056/NEJMsa042830	Yes	Yes	Yes	Yes	Yes	Yes	High	
10	Omer, K.	2008	Cluster RCT	Omer, K., Mhatre, S., Ansari, N., Laucirica, J., & Andersson, N. (2008). Evidence-based training of frontline health workers for door-to-door health promotion: a pilot randomized controlled cluster trial with Lady Health Workers in Sindh Province, Pakistan. Patient Educ Couns, 72(2), 178-185. doi:10.1016/j.pec.2008.02.018	Yes	Yes	Yes	Yes	Yes	Yes	High	

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11	Kumar, R.	2020	Non-randomized quasi-experiment	Kumar, R., Farzeen, M., Hafeez, A., Achakzai, B. K., Vankwani, M., Lal, M., Somrongthong, R. (2020). Effectiveness of a health education intervention on the use of long-lasting insecticidal nets for the prevention of malaria in pregnant women of Pakistan: a quasi-experimental study. Malar J, 19(1), 232. doi:10.1186/s12936-020-03298-2	No	-	-	-	-	Yes	Medium	Union council areas were sampled randomly. However, women were sampled in a non-randomized manner (as many pregnant women did not consent to participate due to cultural or health issues).
12	Hirani, S. S.	2018	RCT	Hirani, S. S., Norris, C. M., Van Vliet, K. J., Van Zanten, S. V., Karmaliani, R., & Lasiuk, G. (2018). Social support intervention to promote resilience and quality of life in women living in Karachi, Pakistan: a randomized controlled trial. Int J Public Health, 63(6), 693-702. doi:10.1007/s00038-018-1101-y	Yes	Yes	Yes	Yes	Yes	Yes	High	
13	Maselko, J	2020	RCT	Maselko, J., Sikander, S., Turner, E. L., Bates, L. M., Ahmad, I., Atif, N., Rahman, A. (2020). Effectiveness of a peer-delivered, psychosocial intervention on maternal depression and child development at 3 years postnatal: a cluster randomised trial in Pakistan. Lancet Psychiatry, 7(9), 775-787. doi:10.1016/s2215-0366(20)30258-3	Yes	Yes	Yes	Yes	Yes	Yes	High	
14	Azmat, S. K.	2013	Non-randomized quasi-experiment	Azmat, S. K., Shaikh, B. T., Hameed, W., Mustafa, G., Hussain, W., Asghar, J., Bilgrami, M. (2013). Impact of social franchising on contraceptive use when complemented by vouchers: a quasi-experimental study in rural Pakistan. PLoS One, 8(9), e74260. doi:10.1371/journal.pone.007426	No	-	-	-	-	Yes	Medium	The control sample could not be randomly selected.
15	Azmat, S. K.	2016	Non-randomized quasi-experiment	Azmat, S. K., Hameed, W., Hamza, H. B., Mustafa, G., Ishaque, M., Abbas, G. Temmerman, M. (2016). Engaging with community-based public and private mid-level providers for promoting the use of modern contraceptive methods in rural Pakistan: results from two innovative birth spacing interventions. Reprod Health, 13, 25. doi:10.1186/s12978-016-0145-9	No	-	-	-	-	Yes	Medium	Non-randomized sample (as two intervention models were being compared- community midwife model vs. voucher scheme)
16	Ali, M	2020	Non-randomized quasi-experiment	Ali, M., Azmat, S. K., Hamza, H. B., & Rahman, M. M. (2020). Assessing Effectiveness of Multipurpose Voucher Scheme to Enhance Contraceptive Choices, Equity, and Child Immunization Coverage: Results of an Interventional Study from Pakistan. Journal of Multidisciplinary Healthcare, 13, 1061.	No	-	-	-	-	Yes	Medium	Union council areas were sampled randomly. Participants were not sampled randomly, but passed on willingness to participate.
17	Midhet, F	2010	RCT	Midhet, F., & Becker, S. (2010). Impact of community-based interventions on maternal and neonatal health indicators: Results from a community randomized trial in rural Balochistan, Pakistan. Reprod Health, 7, 30. doi:10.1186/1742-4755-7-30	Yes	Yes	Yes	Yes	Yes	Yes	High	
18	Qureshi, R. N	2020	RCT	Qureshi, R. N., Sheikh, S., Hoodbhoy, Z., Sharma, S., Vidler, M., Payne, B. A., Bhutta, Z. A. (2020). Community-level interventions for pre-eclampsia (CLIP) in Pakistan: A cluster randomised controlled trial. Pregnancy Hypertens, 22, 109-118. doi:10.1016/j.preghy.2020.07.011	Yes	Yes	Yes	Yes	Yes	Yes	High	

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Supplementary File 3:

Characteristics of Included Studies

Table 1:
Included intervention-based studies

No.	First author	Year	Study Design	Setting	Participants	Sample size	Aim	Type of Intervention	Outcome being measured post intervention	Intervention Delivered by
Counseling Interventions										
1	Ali, B. S	2003	RCT	Semi-urban community in Karachi city	Women aged 18 to 50 years, screened for anxiety and/or depression	Experiment: n= 70 Control: n= 91	Assess if counseling sessions reduce anxiety and/or depression	Mental Health Counseling (one on one)	Reduction in the mean anxiety and depression scores	Briefly trained women counselors of community
2	Khan, M. N.	2017	Cluster RCT	Swat rural district	Women aged 18 or more years affected by conflict	Experiment: n= 59 Control: n= 60	Evaluate the feasibility and acceptability of a group psychotherapeutic interventions, based upon cognitive behavioral approach, to reduce anxiety and depression	Mental Health Counseling (in group)	Decline in psychological distress, measured by levels of anxiety and depression, post-traumatic stress disorder (PTSD), and assessment of general psychological profile and levels of functioning	Briefly trained female non-specialists and LHWs
3	Rahman, A	2019	Cluster RCT	Two rural union councils of Swat city (Odigram and Ghalegay)	Women aged 18±60 years affected by conflict	Experiment: n= 288 Control: n= 290	Effectiveness of a group psychological intervention to improve mental health	Mental Health Counseling (group session)	Reductions in symptoms of anxiety, depression, PTSD, and functional impairment, improved social support, and reduced prevalence of depressive disorders 3 months after the intervention	Briefly trained female non-specialists and LHWs
4	Ali, N. S	2010	Non-randomized quasi-experiment	Qayoomabad and Manzoor Colony of Karachi city	Mothers (with children 0-30 months) screened for postnatal depression	Experiment: n= 62 Control: n= 12	Assess the benefits of counseling on reducing postnatal depression	Mental Health Counseling (one on one)	Reduction in the mean anxiety and depression scores	Women from the community
5	Sikander, S.	2019	Cluster RCT	40 village clusters in Rawalpindi	Pregnant women	Experiment: n= 283 Control: n= 287	Aimed to reduce perinatal depression	Mental Health Counseling (one on one and group)	Severity of depressive symptoms and the prevalence of remission 6 months after childbirth	Volunteer peers and LHWs

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6	Rahman, A.	2008	RCT	40 Union Council clusters in rural Rawalpindi city	Pregnant women	Experiment: n= 463 Control: n= 440	Provide a CBT intervention for mothers with depression	Mental Health Counseling (group session)	Infant weight and height at 6 months and 12 months and prevalence of maternal depression	Trained LHWs
7	Hameed, W	2016	Non-inferiority trial without randomisation	Twenty-two (16 rural- and 6 urban-based) franchised reproductive healthcare facilities in district Chakwal	Married women using long-acting reversible contraceptive	Experiment: n= 1,246 Control: none	Compare the effectiveness of active (doorstep and telephonic) and passive (needs-based response) follow-up (counseling) in sustaining the use of long-acting reversible contraceptive (LARC)	Family Planning Counseling (one on one)	Sustaining the use of long-acting reversible contraceptive	Female Health Educators
8	Sikander, S.	2015	RCT	40 Union Councils in Manshera city	Pregnant women	Experiment: n= 210 Control: n= 211	Assess the effectiveness of cognitive-behavioral counseling on the rate and duration of exclusive breastfeeding (EBF) during the first 6 months of an	Family Planning Counseling (one on one)	Increase in exclusive breastfeeding and decline in use prelacteal feeds	LHWs
Health Education and Awareness Interventions										
1	Jokhio, A. H	2005	Cluster RCT	7 subdistricts Larkana city, Sindh province	Pregnant women	Experiment: n= 10,093 Control: n= 9,432	Provide training and other support to TBAs to reduce perinatal and maternal mortality and reduce complications of pregnancy	Health education and awareness	Decline in perinatal death and maternal mortality	TBAS, LHWs and Obstetricians
2	Omer, K.	2008	Cluster RCT	10 underdeveloped areas from three cities in Sindh (Karachi West, Karachi South and Hyderabad)	Pregnant women	Experiment: n= 529 Control: n= 541	Assess the impact of developing community-based communication tools to promote favorable maternal health practices	Health education and awareness	Improvement in prenatal checkups, care during pregnancy, feeding baby colostrum and maintaining exclusive breastfeeding	LHWs
3	Kumar, R.	2020	Non-randomized quasi-experiment	Two Union Councils (UCs) of Tarparkar district, Sindh	Pregnant women with children up to 6 months of age	Experiment: n= 100 Control: n= 100	Effectiveness of a health education intervention for the prevention of malaria in pregnant women with children up to 6 months of age	Health education and awareness	Knowledge of malaria prevention & use of long-lasting insecticide treated bed nets	LHWs

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				Province						
Social and Psychosocial Interventions										
1	Hirani, S. S.	2018	RCT	Low socioeconomic areas of Karachi city	Young adult women, 20–45 years	Experiment: n= 60 Control: n= 60	Provide a social support intervention to enhance resilience and quality of life	Social, psychosocial or preventive *Social- group sessions*	Improvement in resilience and quality of life	CHWs
2	Maselko, J	2020	RCT	40 village clusters in Kallar Syedan, a rural subdistrict of Rawalpindi city	Pregnant women	Experiment: n= 206 Control: n= 683	Assess the effectiveness of psychosocial intervention on maternal depression and child development (group session)	Social, psychosocial or preventive *Psychosocial-group sessions*	Decline in maternal depression symptoms and remission	Non-specialists peers (lay married women who lived in the same community as that of the women with depression and volunteered their time)
Combination Interventions										
1	Azmat, S. K.	2013	Non-randomized quasi-experiment	Rural and under-served communities of (i) Badin and Dadu-Sindh province, and (ii) Jhang and Khanewal-Punjab province	Married women reproductive age	Experiment: n= 2,483 Control: n= 2,509	Evaluate the effectiveness of a social franchise programme along with a free voucher scheme to promote awareness and use of modern long term contraceptive	Combination	Assess improvement in contraceptive awareness and long-term contraception use	Female health visitors, midwives and nurses at their respective private clinics
2	Azmat, S. K.	2016	Non-randomized quasi-experiment	Eight districts of Sindh, Punjab and KPK	Married women reproductive age	Experiment: n= 1,817 Control: n= 1,075	Provide accessible and affordable long term family planning services through marketing, branding and introducing a voucher scheme	Combination	Increasing awareness about FP methods and use of contraception	Trained facilitators and CMWs
3	Ali, M	2020	Non-randomized quasi-experiment	Faisalabad (urban) & Toba Tek Singh (rural),	Married women of reproductive age	Experiment: n= 1,276 Control: n= 1,276	Assess the effectiveness of a subsidized, multipurpose voucher intervention to enhance the client–provider interaction for improved	Combination	Increase in modern contraception use, knowledge of contraceptives, receiving ANC, and delivery at health	Vouchers and awareness outreach through service providers and distributing

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				Punjab			contraceptive counseling, modern contraception methods uptake, continued use, and its impact on equity through better targeting, while increasing uptake of postnatal care and child immunization among women from the lowest two wealth quintiles.		facilities (also child vaccination rates for BCG, DPT, HBV, and measles)	agencies
4	Midhet, F	2010	RCT	32 village clusters in Khuzdar, a rural district of Baluchistan province	Pregnant women	Experiment: n= 1,539 Control: n= 1,022	Assess the impact of education sessions for safe motherhood practices, TBA training, and emergency transport system on maternal and neonatal health indicators	Combination	Reduction of delays in seeking medical care for obstetric complications and reduction in perinatal and neonatal mortality	Local women with secondary school education or above, & TBAs
5	Qureshi, R. N	2020	RCT	Matiari and Hyderabad districts, Sindh Province	Pregnant women screened for preeclampsia	Experiment: n= 20,264 Control: n= 19,182	Assess the impact of combined interventions for pre-eclampsia (Mobile health assessment; referral; doorstep visits; and educational sessions)to reduce all-cause maternal and perinatal mortality and major morbidity	Combination	Reduction in stillbirths, maternal death or morbidity, early or late neonatal deaths and neonatal morbidity	LHWs

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Page 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Page 1
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Page 4
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Page 4
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Page 4-5
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Page 4-6
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Page 4-6
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Page 4-6
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Page 4-6
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Page 4-6
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Page 4-6
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Page 4-6
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Page 4-6
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Page 4-6
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Page 4-6
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Page 4-6
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Page 4-6
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	n/a
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	n/a
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	n/a
Certainty	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Page 5

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
assessment			
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Page 6
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Page 5-6
Study characteristics	17	Cite each included study and present its characteristics.	Page 8-11
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	n/a
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Page 11-18
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Page 11-18
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Page 11-18
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Page 11-18
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	n/a
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	n/a
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Page 5
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Page 19-22
	23b	Discuss any limitations of the evidence included in the review.	Page 22-23
	23c	Discuss any limitations of the review processes used.	Page 22-23
	23d	Discuss implications of the results for practice, policy, and future research.	Page 23
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Page 2
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Page 2
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	n/a
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page 24
Competing interests	26	Declare any competing interests of review authors.	Page 24
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Page 24

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From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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A Narrative Synthesis Systematic Review of Pakistani Women's Health Outcomes from Primary Care Interventions

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Primary Subject Heading:	Health services research
Secondary Subject Heading:	Health policy
Keywords:	PRIMARY CARE, Health policy < HEALTH SERVICES ADMINISTRATION & MANAGEMENT, Organisation of health services < HEALTH SERVICES ADMINISTRATION & MANAGEMENT

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A Narrative Synthesis Systematic Review of Pakistani Women's Health Outcomes from Primary Care Interventions

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Word Count: 9,989

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ABSTRACT

Objective

Women living in Pakistan have complex health problems including infectious and non-communicable diseases, accident and injuries, and mental health problems. While a majority of these women rely on primary healthcare services for all of their healthcare needs there has to date been no overview of the extent of their effectiveness. The objective of this review was to (i) synthesise the available evidence regarding the effectiveness of primary care based interventions aimed at improving women's mental and physical health; and (ii) identify the factors that promote effectiveness for women's health outcomes.

Methods

Five academic databases were searched, including PubMed, BMC Medicine, Medline, CINAHL and the Cochrane Library. A search was also made of the Grey literature. The quality of included studies was assessed using a standardized critical appraisal tool, and the findings summarized using a narrative synthesis.

Results

In total 18 studies were included in the review. Eight involved evaluations of counseling interventions, three health education and awareness interventions, two social and psychosocial interventions, and five were evaluations of combination interventions. Twelve of the included studies were randomized controlled trials (RCTs). Of these 14 reported significant outcomes, and 4 further interventions showed partially favorable results. However, interventions mostly targeted women's mental or reproductive health.

Conclusions: While the evidence is limited in terms of quality and what has been evaluated, a number of interventions appear to be effective in improving outcomes for women. The three key approaches include the adoption of an active door-to-door and group-based approach; utilization of community peers who can deliver care cost-effectively and who are more accepted in the community; and the integration of financial vouchers to support uptake in poor populations.

PROSPERO registration number CRD42020203472.

PROTOCOL Publication doi:10.1136/bmjopen-2020-043715

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Keywords: Narrative synthesis; systematic review; Pakistan; Women's Health Outcomes; Primary Care Interventions

Strengths and Limitations of this study

- This is the first synthesis of evidence about the effectiveness of primary-care based interventions for women in Pakistan.
- The quality of the included studies was appraised using a standardized tool.
- The diverse nature of the interventions included in this review and assessment of different outcomes meant that it was only possible to provide a narrative synthesis.
- The small number of studies, limited geographical reach, and some studies in the review used patient-reported outcomes only and thereby no objective clinical data, which limits the generalizability of the findings.

Introduction

Creating an effective primary care system for women has not been a priority in Pakistan[1]. One of the difficulties is that a model for an effective primary healthcare setup, for women of different age groups and across the life course has not been agreed upon globally[2], or within Pakistan, to date[3]. Research suggests that for a comprehensive primary healthcare system for women to be effective in developing regions, service delivery must be led by state efforts and public health policy[4]. There is also agreement that a primary healthcare model must be developed regionally, based on a nations historical background, health policies, revenue system, and cultural beliefs, and based on a review of existing services[5].

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Despite the inadequate primary healthcare services and low utilization[6], the majority of Pakistani women need to be provided state support for primary healthcare due to critical socio-economic problems such as poverty and unemployment [7], or informal sector employment which excludes women from employer health protection [8]. There is a need to further support women at the primary care level to reduce maternal mortality [9], chronic disease [10], infectious diseases [11], injury and violence [12], mental health problems [13], and health issues related to aging [14]. The country also has a number of unmet Sustainable Development Goals, with key issues such as high maternal mortality rates and the absence of universal healthcare services for women[15].

Primary healthcare in Pakistan

Since 2011 responsibility for the delivery of Pakistan's healthcare system has been devolved to the provinces. The Federal Ministry of National Health Service Coordination and Regulation is responsible for monitoring and allocation of the budget to the four provinces of the country including Baluchistan, Khyber Pakhtunkhwa, Sindh and Punjab. The overall framework and structure for the delivery of primary healthcare services is the same across the country, with services including: reproductive and maternal health; vaccination and immunisation; health awareness; referral to secondary and tertiary sectors; and routine counseling. However, the outreach and implementation varies according to provincial development, with Punjab and KPK provinces performing better, compared to Baluchistan, Sindh and Gilgit-Baltistan [16].

The current state-provided universal healthcare system in the country, the Sehat Sahulat Program, primarily focuses on payment for hospitalization, with a capping of USD 1,906 per person[17]. This program is severely limited due to low outreach of less than 0.5% of the national population, and is only accessible to women with a national identity card, excluding thereby the majority of disadvantaged women in the country[17]. Pakistan has a functional primary care setup for women,

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called the Lady health Workers (LHW) Programme, which was launched by the government of Pakistan in 1994, and delivers door-to-door services to women at their homes[18]. LHWs provide key services for immunization, family planning, counseling and maternal health services. In addition, there are community midwives (CMWs) trained and then deployed to practice privately in the community for safe delivery services[19]. Primary care services are also provided within the community at basic health units (BHUs) and rural health centers (RHCs), which are allocated a physician or medical officer and a nurse, apart from the LHW. The BHUs and RHCs aim to provide clinical consultancy and basic reproductive health services.

Existing evidence

While a number of studies have reported that the LHW programme and primary setup for maternal healthcare has been successful in improving family planning [20, 21] and mortality rates [22], others have reported that primary health care services have not yielded positive outcomes for women's health in Pakistan [15, 23, 24]. This reflects the fact that access and outreach remains low for the 110 million women of Pakistan, primarily due to the inadequate ratio of primary care services with regard to LHW/CMW to population ratios; BHU and RHC doctor to population ratios; and bed to population ratios for RHCs [25] [26]. The LHW and CMW programs have also been criticised in terms of the limited services for family planning[27]. During the recent pandemic the LHWs were additionally assigned responsibility for promoting infection preparedness and prevention, without any additional compensation or staff allocation, contributing to inefficiencies [28].

Similarly, the BHU and RHC services have also been criticized due to low quality services, inadequate budget allocation, and poor coordination from the center [29]. There is also low

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utilization and visitation to BHUs and RHCs by women due to patriarchal traditions preventing mobility, mistrust, and distance[30]. To date there has been no systematic review of rigorous evaluations to identify which primary care interventions are effective in improving women's health outcomes in Pakistan. The aim of this review was to systematically review the available literature for studies that have evaluated the effectiveness of primary healthcare services for women, in order to inform future health research and policy.

METHODS

Patient and public involvement

No patients are involved in this study.

Search strategy

A systematic review using a narrative synthesis was undertaken. Electronic databases including Pubmed, BMC, Medline, CINAHL, and Cochrane Library were searched. All published data prior to October 2020 was searched. We started the search at the beginning of August 2020 and finished it as the end of October 2020. We were unable to start searching during the dates mentioned in the protocol (January to June 2021) due to a combination of factors such as investigator time constraints, transfer to online teaching for faculty, and COVID-19 social distancing making it difficult for co-authors to meet and coordinate. A search of the grey literature included the following databases: Google, Google Scholar, WHO website and Government of Pakistan Health Ministry websites. Detailed information about the search strategy, search terms, inclusion and exclusion criterion, and methods for this study can be found in the protocol publication [31]. A summary of the search terms and database search results can be found in **Supplementary File 1.**

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The inclusion criteria were established using PICOS: (i) Population: The population of interest was all Pakistani women, above 18 years of age, and excluding children and adolescent girls; (ii) Intervention: all healthcare interventions that are delivered in the primary healthcare sector; (iii) Comparator: the review only included studies that used a control or comparison group; (iv) Outcomes: All studies that focused on physical or mental health outcomes, were included in the review; (v) Study design: all randomized controlled trials (RCTs) and quasi-experimental study designs that involved the use of a control group. The language was restricted to English.

Initial Screening

An initial screening of studies based on titles, was performed by the first author for 109,841 studies. An independent screening was also done by the second author. Discussions about which articles met the inclusion criterion were held. At second step, the abstracts of 543 studies were reviewed by the first and third authors independently. The main reasons for exclusion were that articles were not interventions in Pakistan or exclusively in Pakistan, did not relate to health interventions or primary health care outcomes for women, were protocol or feasibility studies, or did not have a control group. The final 18 included articles had a diverse range of interventions and outcomes, which meant that it was only possible to provide a narrative synthesis32.

Although the protocol refers to two objectives (the primary objective is to identify the effectiveness of the intervention in terms of how successful it was in improving health of women; whereas the secondary aim is to identify barriers and facilitators for delivery of primary healthcare services) this paper only reports the findings of the first of these. The review pertaining to the secondary aim referred to in the protocol paper, will be submitted as a separate manuscript entitled: Barriers and facilitators to the delivery of primary health care services for women in Pakistan.

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Data extraction

The following information was abstracted, by the first author, from the interventions that met our eligibility criteria in terms of targeting women's health outcomes: study design; setting; participant type; sample size; aim; type of intervention; intervention delivery method and description; outcomes being measured; data analysis technique; results; and complete citation of the study. Extraction of the data was reviewed for consistency by the third author.

Critical appraisal

The included studies were appraised using standardized critical appraisal tools. The RCTs were appraised using the RCT Critical Appraisal Tool [32], and the non-randomized studies were appraised using the Mixed Methods Appraisal Tool (MMAT)[33]. Both appraisal tools score rate studies as being of High, Medium, or Low quality. The quality of the articles was assessed by two reviewers (SRJ and QKM), and group meetings were held to discuss the independent appraisal of articles and to resolve discrepancies. Twelve of the eighteen included articles (all the RCTs) scored high on the six evaluation areas of the critical appraisal checklist. Six studies were rated as medium quality due to the failure to use randomization and inability to administer the intervention as intended. No studies received a low quality rating or were excluded from the review based on the quality assessment. More information related to the critical appraisal for each study, the description of assessment criterion and the scoring for the included articles in this review has been included in Supplementary File 2.

Data management and synthesis

Studies from databases were transferred into Endnote Software by the first author. After the exclusion of duplicates, studies were transferred to an Excel template for data extraction and

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organization. The third author reviewed this extraction and checked for consistency. Final selection of studies to be included was done independently by the first two authors. Separate tabs were then created for each study for detailed data extraction and summary sheets were also created to organize study characteristics and information for supplementary data files. Due to the high level of heterogeneity across studies in terms of interventions and outcomes, a narrative summary of the findings was necessary.

RESULTS

Included Studies

The search results have been reported using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses flow chart, presented in **Figure 1**. A total of 135,553 publications were identified, of which 5,405 were duplicates. After duplicates were removed 109,841 were selected for further screening of titles. Following a review of the abstracts, the full text of 543 publications were retrieved, and assessed against the selection criteria, and a total of 18 studies were included of interventions assessing the effectiveness of primary healthcare interventions for women's health outcomes. **Table 1** lists the interventions including information about the authors, study type, sample, description of intervention, and post-intervention outcomes with p-values. Further details of the characteristics of included studies have been reported in **Supplementary File 3**.

Study characteristics

The interventions included in this review (**Table 1**) were classified as follows: (i) Eight studies were classified as 'Counseling Interventions' [21, 34-41]; (ii) three as 'Health Education and Awareness Interventions' [42-45]; (iii) two as 'Social and Psychosocial Interventions' [46-48];

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and (iv) five as 'Combination Interventions (e.g. *vouchers, education, referral and technology)* [49-55].

With regard to the study designs: (a) 12 studies are RCTs (including 7 RCTs and 5 cluster RCTs); and (b) 6 are quasi-experimental studies including 1 non-randomized non-inferiority trial. A majority of the studies (n=9) delivered interventions to pregnant women. Another 5 studies delivered interventions to women of reproductive years. Two studies targeted women aged 18 and above affected by conflict and another 2 delivered the intervention to women screened for depression. Overall therefore, a majority of included studies targeted health outcomes for pregnant women or women of reproductive years.

Nearly all the interventions (n=15) used patient-reported outcomes, with 3 reporting both patient-reported and clinical outcomes. None of the interventions included a sample from across all the provinces of Pakistan. Six of the interventions were limited to only one city or town, 10 to one province of Pakistan, and 2 studies included samples from two or three provinces in the country. A small number of interventions were delivered exclusively by lady health workers (LHWs) (n=5), and another 6 interventions were facilitated by a combination of LHWs, community health workers (CHWs), community midwives (CMWs), traditional birth attendants, and laywomen. The remaining interventions were delivered by either laywomen or local peers (n=3); trained healthcare staff, nurses or health educators (n=3); or by CHWs alone (n=1). Fourteen of the included studies showed significant results for primary outcomes and 4 showed partially favorable results.

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Table 1:
List of included intervention-based studies to review effectiveness on women's health outcomes

Authors	Study Type	Sample	Intervention	Results (p value)
Ali, B. S., Rahbar, M. H., Naeem, S., Gul, A., Mubeen, S., & Iqbal, A. (2003)	RCT	Women aged 18 to 50 years screened for anxiety or depression (Experiment: n= 70; Control: n= 91)	One-on-one counseling sessions to reduce anxiety and/or depression	A significant reduction was found between the mean anxiety and depression scores of both the intervention and control groups (p = 0.000)
Khan, M. N., Hamdani, S. U., Chiumento, A., Dawson, K., Bryant, R. A., Sijbrandij, M., Rahman, A. (2017)	Cluster RCT	Women aged 18 or more years affected by conflict (Experiment: n= 59 Control: n= 60)	Group psychotherapeutic intervention to reduce anxiety and depression in women	Statistically significant decline in PTSD symptoms (p=<0.0001) and depressive disorder scores (p= 0.0016) of the intervention compared to the control group
Rahman, A., Khan, M. N., Hamdani, S. U., Chiumento, A., Akhtar, P., Nazir, H., van Ommeren, M. (2019)	Cluster RCT	Women aged 18–60 years affected by conflict (Experiment: n= 288 Control: n= 290)	Group psychological intervention to improve mental health	A significant lower mean total score of anxiety and depression in women in the intervention group compared to the control group (p=0·0007)
Ali, N. S., Ali, B. S., Azam, I. S., & Khuwaja, A. K. (2010)	Non-randomized quasi-experiment	Mothers, with children 0-30 months, screened for postnatal depression (Experiment: n= 62 Control: n= 12)	One-on-one counseling to reduce postnatal depression	A significant decline in level of anxiety/depression was found in both the counseled and the non-counseled groups (p= < 0.001); with a greater decline in the counseled/ intervention group. The counseled group fared better than the non-counseled for (i) recovery; (ii) reduction in the rate of recurrence; and (iii) increase in the duration before relapse.
Sikander, S., Ahmad, I., Atif, N., Zaidi, A., Vanobberghen, F., Weiss, H. A., Rahman, A. (2019)	Cluster RCT	Pregnant women (Experiment: n= 283 Control: n= 287)	One-on-one and group counseling to reduce perinatal depression	The depression mean scores (p=0·07) and the prevalence of remission (p=0·14) did not significantly differ between the intervention and control groups at 6 months. However, the intervention group, compared to the control group, reported: (i) reduced depression symptom severity (p<0.0001) and disability score (p<0.0001) at 3 months; (ii) more likely remission (p=0.002) and recovery (p=0.002) at 3 months; and (iii) improvement in perceived social support (p=0.007) at 6 months after childbirth.
Rahman, A., Malik, A., Sikander, S., Roberts, C., & Creed, F. (2008)	RCT	Pregnant women (Experiment: n= 463 Control: n= 440)	Group cognitive behavioral therapy for mothers with depression	The intervention group had lower depression scores at two time points compared to the control group: 1. 6 month point decline (p-value= <0·0001) 2. 12 month point decline (p=<0·0001). Intervention group also showed significant improvement at both time points for (i) overall functioning (6 months= <0.0001; 12 months= <0•0001) and (ii) perception of social support (6 months= <0.0001; 12 months= <0.0001).
Hameed, W., Azmat, S. K., Ali, M., Ishaque, M., Abbas, G., Munroe, E., Ahmed, A. (2016)	Non-inferiority trial without randomisation	Married women using long-acting reversible contraceptive (Experiment: n= 1,246 Control: none)	Compare active (doorstep and telephonic) versus passive (needs-based) follow-up in sustaining use of long-acting reversible contraceptive	Active follow-up is more effective (compared to passive approach) in sustaining long acting reversible contraceptives than the passive (needs-based) follow-up (p= 0.035). Telephone-based follow-up is as effective as the home-based follow-up (compared to passive approach) in sustaining long acting reversible contraceptives (p= 0.431)
Sikander, S., Maselko, J., Zafar, S., Haq, Z., Ahmad, I., Ahmad, M., Rahman, A. (2015)	RCT	Pregnant women (Experiment: n= 210 Control: n= 211)	Cognitive-behavioral counseling to improve rate and duration of exclusive breastfeeding during first 6 months	Increase in exclusive breastfeeding (p = < 0.001) and decline in the use prelacteal feeds (p = < 0.001) in the intervention group over the control group.
Jokhio, A. H., Winter, H. R., & Cheng, K. K. (2005)	Cluster RCT	Pregnant women (Experiment: n= 10,093 Control: n= 9,432)	Training to TBAs to reduce perinatal and maternal mortality and	The intervention areas, compared to the control, had lower odds of: (i) perinatal death (p=<0.001) and (ii) maternal mortality of (p=<0.001).

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			reduce complications of pregnancy	
Omer, K., Mhatre, S., Ansari, N., Laucirica, J., & Andersson, N. (2008)	Cluster RCT	Pregnant women (Experiment: n= 529 Control: n= 541)	Developing community-based communication tools to promote favorable maternal health practices	Women in the intervention communities were more likely to attend prenatal checkups, to stop routine heavy work during pregnancy, to give colostrum to newborn babies, and to maintain exclusive breastfeeding for four months. No p values were reported.
Kumar, R., Farzeen, M., Hafeez, A., Achakzai, B. K., Vankwani, M., Lal, M., Somrongthong, R. (2020)	Non-randomized quasi-experiment	Pregnant women with children up to 6 months of age (Experiment: n= 100 Control: n= 100)	Health education intervention for the prevention of malaria in pregnant women with children up to 6 months of age	The intervention group, compared to the control, showed an increase in: (i) knowledge score (p= < 0.01), and (ii) use of long-lasting insecticide treated bed nets (p < 0.05).
Hirani, S. S., Norris, C. M., Van Vliet, K. J., Van Zanten, S. V., Karmaliani, R., & Lasiuk, G. (2018)	RCT	Young adult women, 20–45 years (Experiment: n= 60 Control: n= 60)	Social support intervention, in groups, to enhance resilience and quality of life	Women in the intervention group reported improvements in: (i) resilience scale-14 item (p = 0.022); (ii) the resilience scale for adults (p = 0.043). No significant findings were noted in the quality of life scores.
Maselko, J., Sikander, S., Turner, E. L., Bates, L. M., Ahmad, I., Atif, N., Rahman, A. (2020)	RCT	Pregnant women (Experiment: n= 206 Control: n= 683)	Group psychosocial intervention to reduce maternal depression	Reduced symptom severity and high remission rates were seen across both the intervention and the control group. Significant outcome differences between the intervention and control group were not found.
Azmat, S. K., Shaikh, B. T., Hameed, W., Mustafa, G., Hussain, W., Asghar, J., Bilgrami, M. (2013)	Non-randomized quasi-experiment	Married women reproductive age (Experiment: n= 2,483 Control: n= 2,509)	Social franchise programme + a free voucher scheme to promote awareness and use of modern long term contraceptive	Improvement in intervention group over the control group in: 1. Awareness of contraception (p < 0.001); 2. Ever use of contraception (p <0.001); 3. Ever use of any modern method (p < 0.0010); 4. Overall contraceptive prevalence rate (p <0.001).
Azmat, S. K., Hameed, W., Hamza, H. B., Mustafa, G., Ishaque, M., Abbas, G., Temmerman, M. (2016)	Non-randomized quasi-experiment	Married women reproductive age (Experiment: n= 1,817 Control: n= 1,075)	Compare two interventions to improve the use of modern contraceptive methods: 1) Social franchise model along with free vouchers and 2) Community Midwife Model	- Both the models were effective for the intervention group over the control group. The Social franchising and free voucher model: (i) contraceptive awareness(p < 0.001), (ii) use of contraceptives (p=<0.0001), (iii) long term modern method-intrauterine device (IUD) use (<0.0001). The Community midwife model: (i) contraceptive awareness (p < 0.001), (ii) use of contraceptives (p=<0.0001) and, (iii) IUD use (0.0078). The social franchising and voucher approach led to a 35 % greater prevalence of contraceptive use in women compared to the CMW model.
Ali, M., Azmat, S. K., Hamza, H. B., & Rahman, M. M. (2020)	Non-randomized quasi-experiment	Married women reproductive age (Experiment: n= 1,276 Control: n= 1,276)	Subsidized multipurpose voucher + family planning counseling to improve contraceptive use, postnatal care and child immunization	There was no increase in modern contraception use in the intervention area (p<0.0001). However, the concentration index and slope index of inequalities for first-time use of modern contraceptives, knowledge of contraceptives, receiving ANC, and delivery at health facilities were negative, indicating that the use of these services was more concentrated among the disadvantaged in intervention areas.
Midhet, F., & Becker, S. (2010)	RCT	Pregnant women (Experiment: n= 1,539 Control: n= 1,022)	Education sessions for safe motherhood practices + TBA training + Emergency transport system to improve maternal and neonatal health indicators	The intervention clusters received prenatal care and prophylactic iron therapy more frequently than pregnant women in control clusters (p=<0.05). Providing safe motherhood education resulted in significant increase in percent of hospital deliveries (p=<0.05) and reduction in perinatal mortality (p=<0.05). There was no impact on the use of skilled birth attendants.

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Qureshi, R. N., Sheikh, S., Hoodbhoy, Z., Sharma, S., Vidler, M., Payne, B. A., Bhutta, Z. A. (2020)	RCT	Pregnant women screened for preeclampsia (Experiment: n= 1,276 Control: n= 1,276)	Mobile health assessment+ referral+ doorstep + educational sessions to reduce all-cause maternal & perinatal mortality & major morbidity	There was no difference between the groups in the primary outcome of composite maternal, fetal and newborn mortality and major morbidity (p= 0·31). However, there was a reduction in stillbirths (p = 0·03) and no adverse events were reported in the intervention group.

Effectiveness of primary healthcare interventions

Due to the heterogeneity in terms of the included study designs and interventions, a narrative report of effectiveness has been provided, organized according to the type of intervention as follows: A. Counseling Interventions- including one-to-one or group sessions to improve mental health or family planning practices; B. Health Education and Awareness Interventions- including group or one-on-one educational sessions to TBAs and women of reproductive years about safe maternal practices or malaria prevention; C. Social and Psychosocial Interventions- including group social support sessions (peer-support, behavioral activation, and problem solving) and one-on-one mental health support; and D. Combination Interventions- Group or one-on-one sessions for family planning and contraception use or wider maternal health indicators using a combination of different approaches such as educational outreach, clinical assessment, transport provision and referral.

A. Counseling Interventions

Eight included studies evaluated the effectiveness of counseling interventions. Six of these interventions targeted improvement in mental health; and two aimed to improve family planning or breastfeeding practices.

i) Mental health

Only one of the six mental health interventions aimed to deliver counseling to all women of any age who had screened positive for anxiety and/or depression. This RCT was conducted with

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women of a semi-urban community in Karachi city and the intervention consisted of one-to-one counseling sessions delivered by briefly trained women from the community [34]. The women in the intervention group received eight sessions of counseling, which included supportive, problem-solving, and cognitive-behavioral techniques. The results showed a significant reduction in both the experimental and control groups post intervention, this possibly having been due to some women in the control group also receiving counseling.

Two studies involved the delivery of services to women affected by conflict in the Swat district, and who screened positive for mental health disorders [36, 41]. Both interventions were group-based programs, involving adaptions of the World Health Organization Problem Management intervention (PM+), and delivered by briefly trained female non-specialists and LHWs. Both cluster RCTs reported a significant decline in anxiety and depression post intervention.

One RCT involved the deliver of one-to-one counseling sessions to mothers with children between 0-30 months, screened for postnatal depression [35]. This non-randomized quasi-experimental study included women from two underdeveloped colonies of Karachi city and the one-to-one counseling was delivered by briefly trained women from the community. The intervention consisted of supportive, problem-solving and basic cognitive behavioral techniques delivered for eight weeks in weekly one-hour sessions. The study found a significant decline in anxiety and depression in both the counseled and the non-counseled groups. This may be because the control group received enhanced social support through regular visits by the counselors although the number of visits was not described. In addition, the study reported a greater decline in the counseled group, compared to the non-counseled for: (i) recovery; (ii) reduction in the rate of recurrence; and (iii) increase in the duration before relapse.

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Two mental health intervention studies evaluated the effectiveness of cognitive behaviour therapy (CBT) in supporting women experiencing maternal depression. In a cluster RCT, involving 40 villages from a rural district of Rawalpindi city, pregnant women who screened for positive for depression were recruited to the trial [39]. The Thinking Healthy Programme (THP) for perinatal depression was used, which includes a combination of strategies that incorporate behavioural activation, active listening, collaboration with the family, guided discovery, and homework, and was delivered by briefly trained volunteer peers from the community with LHWs as facilitators. The intervention consisted of ten individual and four group sessions, from the third trimester of pregnancy to 6 months after childbirth. The intervention group, compared to the control group, reported significant improvements in: (i) depression symptom severity and disability score at 3 months; (ii) remission and recovery at 3 months; and (iii) perceived social support at 6 months after childbirth.

The second RCT involved the delivery of CBT to women from 40 villages from the rural district of Rawalpindi city [21]. Pregnant women screened for depression received the THP intervention, in group sessions, delivered by specially trained LHWs. Significantly lower depression scores were reported for the intervention group, at the 6- and 12-month time points. Additionally, mothers in the intervention group showed significant improvement in overall functioning and perception of social support, at both 6 and 12 month time points. Both studies were found to be cost-effective for resource-poor nations as they utilized volunteer peers to deliver the intervention.

ii) Family planning practices

Two included studies used counseling to improve family planning practices. In one non-inferiority trial without randomisation, a comparison was made between active and passive

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follow-up of existing long-acting reversible contraceptive-use (LARC) [38]. Married women using long-acting reversible contraception were recruited from twenty-two rural and urban primary healthcare facilities of Chakwal city. Female health educators delivered one-to-one sessions using an adapted manual for family planning services based on the Family Advancement for Life and Health Project. Results showed that both the telephone-based follow up and home-based visits significantly improved the use of LARC, compared to the passive or needs-based follow-up.

The second study, an RCT, evaluated the effectiveness of an intervention to improve temporary contraceptive methods through breast-feeding practices. The intervention was delivered by LHWs using one-to-one cognitive-behavioral counseling sessions aimed at improving the rate and duration of exclusive breastfeeding during the first 6 months of an infant's life [40]. The results showed a significant increase in the duration of exclusive breast-feeding and also in rates of exclusive breast-feeding at six months postpartum in the intervention group. The study also reported a decline in the use of prelacteal feeds to infants in the intervention group.

B. Health Education and Awareness Interventions

Three studies evaluated the effectiveness of health education and awareness interventions. Two of these involved the use of cluster RCTs to improve maternal health indicators and practices. One cluster RCT involved training traditional birth attendants (TBAs), and also providing them with safe delivery kits, across seven sub-districts in Larkana city [44]. A team of obstetricians and female paramedics trained the TBAs; and LHWs were further trained to provide support to the TBAs in the community. The training involved the use of picture cards about safe practices for antepartum, intrapartum, and postpartum care. Additionally, they were trained in how to

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conduct a clean delivery; use of the disposable delivery kit; and when to refer women for emergency obstetrical care. The results showed significantly lower odds of perinatal death and maternal mortality in the intervention group.

A second cluster RCT involved the delivery of educational sessions to pregnant women or mothers with young children about safe maternal practices [45]. Women from ten underdeveloped areas from three cities of Sindh province were included and the awareness communication tool was developed in consultation sessions between LHWs and government health officials. It was decided that embroidered pictures on a traditional Sindh fabric (known as *ajrak*) should be used to communicate user-friendly, indigenous, and attractive messages to raise awareness about the importance of antenatal check-ups, giving colostrum after birth, and not doing heavy work during pregnancy. The intervention group showed an increase in prenatal checkups, stopping routine heavy work during pregnancy, provision of colostrum to newborns, and exclusive breastfeeding.

The third study evaluated the effectiveness of a health education and awareness intervention aimed at improving malaria net use in pregnant women [42]. This non-randomized quasi-experimental study included women from two areas of Tarparkar district. LHWs were responsible for providing women with information about malaria prevention and the importance of using long-lasting in-bed nets (LLIN). Results showed a significant increase in the experimental group's knowledge and use of LLIN.

C. Social and Psychosocial Interventions

Two RCTs involved an evaluation of the effectiveness of interventions providing social or psychosocial support to improve resilience and quality of life, and to reduce maternal depression,

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respectively. The first study was delivered to young adult women, 20-45 years, in low socioeconomic areas of Karachi city [46]. Women were provided with social support in six weekly group sessions, using a manual that guided women about stress management and coping strategies for stress. Results revealed a statically significant improvement in resilience for the intervention group, but no impact on the quality of life (QOL) indicators.

The second study, assessed the effectiveness of a psychosocial intervention in pregnant women screened for depression [48]. The intervention was delivered across forty village clusters of Rawalpindi city by non-specialist volunteer peers who were lay married women living in the same community. Both the intervention and control groups received enhanced usual care from the LHW program, which consisted of: (i) informing participants about their depression status and ways to seek help for it and also providing them with an information leaflet, (ii) informing LHWs about each woman's depression status, and (iii) training the primary care facility-based physicians in the sub-district in the management of maternal depression. In addition, the intervention group received a longer-duration, psychosocial intervention, consisting of 18 group-based sessions (from 7 months to 36 months postnatal) that included peer-support, behavioral activation, and problem solving in a culturally sensitive and non-medicalised format.
The findings showed a decline in maternal depression symptoms and remission in both the intervention and enhanced care groups, thereby masking any potential benefits of the intervention.

D. Combination Interventions

Five studies evaluated the effectiveness of combination interventions on a) family planning and contraception use; or b) wider maternal health indicators. The combined interventions included:

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(i) Voucher system *plus* educational outreach, (ii) Education sessions for safe motherhood practices, *plus* TBA training, *plus* emergency transport system; and (iii) Mobile health-guided clinical assessment, *plus* referral to facility for specialized treatment, *plus* educational sessions.

i) Family planning and contraception use

Three studies evaluated the effectiveness of family planning and contraceptive use. The first of these was a quasi-experimental study, which included married women of reproductive age from rural and under-served communities of two provinces - Punjab and Sindh [50]. A social franchise programme delivered by female health visitors, midwives and nurses, along with a free voucher scheme, was used to promote awareness and use of modern long-term contraceptives. Results showed a statistically significant improvement in the intervention group for (i) awareness of modern contraception, (ii) ever use of modern contraceptives, (iii) overall contraceptive prevalence rate, and (iv) uptake of intrauterine devices.

A second quasi-experimental study compared a social franchising and voucher model with educational sessions provided by CMWs [51]. The study included married women of reproductive age from eight districts of three provinces - Sindh, Punjab and KPK. Trained facilitators and CMWs delivered the intervention and results showed a significant increase in contraception awareness, contraception use, and long-term modern method-intrauterine device through both the social franchising and voucher approach and the CMW model. However, the social franchising and voucher approach led to a 35% greater uptake of contraceptive use.

A third non-randomized quasi-experimental study assessed the effectiveness of a subsidized multipurpose voucher *plus* family planning counseling to improve contraceptive use and postnatal care [55]. Married women of reproductive age were included from one city each from urban and rural Punjab. The intervention was delivered by community service providers

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who were trained to provide family planning counseling and supported by local distributing agencies who provided vouchers. Although there was no net increase in modern contraception use there was a low discontinuation rate, and vaccination rates for BCG increased significantly by 14%, and 5% each for DPT, HBV, and measles. Furthermore, use of services was more concentrated among the disadvantaged in intervention areas than the wealthy counterparts, thereby suggesting increased equity.

ii) Maternal health indictors

Two RCTs evaluated the effectiveness of multicomponent interventions on maternal health indicators. The first RCT evaluated the effectiveness of education sessions promoting safe motherhood practices *plus* TBA training *plus* an emergency transport system [53]. Both pregnant women and their husbands received the intervention across thirty-two village clusters of Khuzdar city, through the combined efforts of local women and TBAs. Results showed a significant increase in hospital deliveries and a reduction in perinatal mortality in the intervention groups. Additionally, the intervention clusters received prenatal care and prophylactic iron therapy more frequently than pregnant women in the control clusters.

The second RCT involved an evaluation of the effectiveness of mobile health-guided clinical assessment *plus* referral to a facility for specialized treatment *plus* educational sessions [54]. Pregnant women screened for preeclampsia were recruited from two cities of the Sindh province. LHWs delivered the intervention and received support from nurses and doctors at primary health centers as and when referrals were made based on need. While the results showed no impact on the composite measure of maternal health indicators, other benefits of the

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intervention were found, such as a statistically significant reduction in stillbirths, and no adverse events were recorded for the intervention group.

DISCUSSION

This is the first systematic review to synthesise evaluations of primary care interventions that are aimed at improving women's health outcomes. Such reviews are essential to the development of effective healthcare policy especially in developing regions [2]. Furthermore, country-specific systematic reviews of this nature are best able to identify regional effectiveness that is central to guiding the developing of such health policy [5]. Four types of intervention were identified by this review- Counseling; Health Education and Awareness; Social and Psychosocial; and Combination Interventions (i.e. vouchers, education, referral and technology) – and all included studies reported favorable or partially favorable results (i.e. on secondary outcomes) pointing to the value of providing such care in terms of improving the health of women in Pakistan. However, most of the included studies evaluated the effectiveness of interventions to women of reproductive years and only four studies evaluated interventions that were not targeting reproductive health or family planning. These findings are consistent with the wider literature, which indicates that interventions for maternal health are given comparative priority, as opposed to health services for other groups such as unmarried women, aging women, and those with special needs or disabilities [56]. In addition, a large number of health conditions appear not to be addressed as part of primary care interventions including (i) chronic disease and multimorbidity [57], and (ii) violence and injury [58]. While the focus of primary healthcare services on women's reproductive health in developing countries and conservative climates may reflect the high maternal and child mortality that is present [59], the failure to address the wider

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health care needs of Pakistan's population of women, leaves a significant health burden both for the women themselves and society more widely.

With regard to the effectiveness of counseling interventions, nine studies showed evidence of effectiveness in terms of their primary goal of improving mental health, and family planning or maternal health practices. Both one-to-one and group-based counseling sessions were successful in improving women's health outcomes, and the results also showed that an active home or telephone-based follow-up was more effective compared to a passive needs-based approach [38], in terms of reinforcing and sustaining behavioural change [60], which is important in a context where a majority of the population are disadvantaged and semi-literate.

The studies of counseling interventions also demonstrated the effectiveness of using low-cost and more acceptable community peers as intervention facilitators, both important points for a culturally conservative and resource poor country like Pakistan [61]. Furthermore, there were also benefits of awareness building and social support as part of routine visits, which was especially valuable for disadvantaged and semi-literate women who needed reaffirmation and regular care contact [62]. It is noteworthy that two studies in this review showed improvement in the mental health of Pakistani women impacted by conflict. Given that a large proportion of people in Pakistan are affected by conflict and natural disasters [63], these findings suggest that such interventions should be integrated into routine primary care services.

The three health education and awareness interventions reported effectiveness in improving preventive practices for Malaria, training of existing TBAs, and the development of community-based tools to advance maternal health practices. Training of TBAs and development of culturally acceptable tools is important in terms of the acceptance and uptake of health services for women in a conservative region like Pakistan. In Pakistani society both culture and

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tradition compel women to seek health services from TBAs as the first line of support [64], and as such, the training of TBAs provides an important way of improving the reproductive of health of Pakistani women.

The psychosocial interventions were found to improve resilience and quality of life for women, which is important in a patriarchal culture where women face barriers to mobility and access to social support [65]. Preventive health care of this nature is critical in terms of improving the wellbeing of women and reducing the national health burden [66]. The group psychosocial intervention served as an effective way of utilizing peer-support and problem-solving in a culturally sensitive and non-medicalised format. Wider research confirms that both group sessions and problem-solving are valuable tools to improve health behaviors and preventive practices in women [67].

All five combination interventions were focused on reproductive health and family planning. The findings from this group of interventions suggest that educational counseling and door-to-door services need to be coupled with financial interventions, such as free or subsidized voucher schemes. Despite the provision of free community services, optimal healthcare outcomes are only possible when financial support is available for transport, private consultancy, and purchase of medicines or other health commodities [68]. For low income nations such as Pakistan, free or subsidized vouchers are integral to effective uptake of services and better health outcomes [49, 51]. Though education, awareness and counseling are effective interventions, many poor women are prevented from using them without financial subsidies [50, 55] and the provision of transport services [53].

One combination intervention highlighted the effectiveness of delivering awareness and training to husbands, which is important to enable women in the region to access services, travel

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for services, and return for continued care. The patriarchal climate of the region, with men as primary decision-makers for women's health behavior and mobility, makes interventions with husbands more effective for women's health outcomes [69].

Overall, the results support the use of (i) an active approach, (ii) delivery by community peers, and (iii) financial vouchers. A number of studies showed that an active approach involving door-to-door visitation [38, 52], is also helpful for building general health awareness and social support [35, 46] and improving preventive health [42, 43]. The training of local peers also appears to be a cost-effective and highly acceptable way to support women living in conservative communities [21, 34, 39, 44, 53]. Furthermore, the evidence suggests that locally trained community women can be even more successful in improving health outcomes when they are partnered with existing government deployed LHWs [36, 41]. However, employing briefly trained non-specialists from the community requires fidelity measures and monitoring to ensure effective and consistent delivery of services [37, 48].

Limitations

The diverse nature of the interventions included in this review, alongside the use of different outcomes makes a comparison of the included interventions difficult. Furthermore, we included seven studies that were not RCTs, and as such cannot be confident that the results of these studies do not reflect a regression to the mean. Overall, the quality of the included studies was good to moderate. The assessment of outcome for all included studies relied on the use of patient-reported outcomes, which is of concern especially in underdeveloped regions with illiterate and semi-literate populations who may not answer the survey or self-report correctly, even with the assistance of data collectors, due to low levels of understanding [70]. It is now recognised that health intervention assessments must combine patient-reported outcomes with

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clinical outcomes in order to provide more robust evidence about health outcomes in women [71].

Most of the studies included in this review were conducted in either one city or one province, and as a result of the fact that the cultural climate and behavioral response for each province of Pakistan is unique [72], it is not possible to make generalizations about the effectiveness of interventions across Pakistan based on the current evidence. Only when interventions are delivered across provinces and remote locations will it be possible to generalize the findings. This will also ensure that the designing of services and policies are ethnically and regionally appropriate and based on the different needs of diverse population groups [73]. As discussed earlier, the studies focused largely on women of reproductive age or pregnant women and the interventions are limited to the delivery of services for family planning and maternal health.

Conclusion

Efforts to enhance the nature and quality of women's primary healthcare services in Pakistan must be informed by contextualized evidence. This systematic review found that to date, evidence regarding the effectiveness of primary care services for women in Pakistan remains limited and focused primarily on reproductive or mental health in only one or two provinces. There is, as such, an absence of information about the effectiveness of primary care services for older women, women with special needs, refugees and displaced women, and unmarried women. This review has highlighted the effectiveness of a number of primary care based services and the benefit of active door-to-door and group-based approaches; utilization of community peers who

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can deliver care cost-effectively and who are more accepted in the community; and the benefit of integrating financial vouchers to support uptake in poor populations.

Contribution Statement

SRJ and JB conceptualized the study, designed the methodology, and wrote the manuscript. SRJ, QKM, SM, and MA performed the literature search. SRJ interpreted the data and JB contributed to the interpretation. JB supervised the project.

Competing Interests

There are no competing interests to declare.

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Data sharing statement

Data sharing not applicable as no datasets generated and/or analysed for this study. All data relevant to the study are included in the article or uploaded as online supplemental information.

Patient and public involvement

No patient involved

Ethics approval

This study does not involve human participants. It is a systematic review of studies and does not require ethics approval from an IRB.

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Supplementary Files:

Supplementary File 1: Search Terms and Database Results
Supplementary File 2: Quality Appraisal of Included Studies
Supplementary File 3: Characteristics of Included Studies

Figure Legend

Figure 1- Preferred Reporting Items for Systematic Reviews and Meta-Analyses flow chart.

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REFERENCES

1. Wazir, M.S., B.T. Shaikh, and A. Ahmed, *National program for family planning and primary health care Pakistan: a SWOT analysis.* J Reproductive health, 2013. **10**(1): p. 1-7.
2. Kringos, D.S., et al., *The breadth of primary care: a systematic literature review of its core dimensions.* J BMC health services research, 2010. **10**(1): p. 1-13.
3. Nasreen, S. and A. Ali, *A Gender Analysis Of Health Policy In Pakistan. Pakistan Journal of Applied Social Sciences.* 4(1), 111-121, 2016.
4. Labonte, R., et al., *Is the Alma Ata vision of comprehensive primary health care viable? Findings from an international project.* J Global health action, 2014. **7**(1): p. 24997.
5. Politzer, R., et al., *Primary care quality: community health center and health maintenance organization.* J Southern medical journal, 2003. **96**(8): p. 787-795.
6. Nishtar, S., et al., *Pakistan's health system: performance and prospects after the 18th Constitutional Amendment.* J The Lancet, 2013. **381**(9884): p. 2193-2206.
7. World Bank, *Pakistan Development Update.* 2017.
8. United Nations Pakistan, *One United Nations Programme III 2018-2022, United Nations Sustainable Development Framework for Pakistan.* 2018.
9. Aziz, A., et al., *Why are the Pakistani maternal, fetal and newborn outcomes so poor compared to other low and middle-income countries?* J Reproductive Health, 2020. **17**(3): p. 1-12.
10. Jafree, S.R., *Determinants of depression in women with chronic disease: Evidence from a sample of poor loan takers from Pakistan.* J Journal of Community Psychology, 2020. **48**(7): p. 2238-2251.
11. Butt, M., et al., *Why have immunization efforts in Pakistan failed to achieve global standards of vaccination uptake and infectious disease control?* J Risk management healthcare policy, 2020. **13**: p. 111.
12. Ali, P.A. and M.I.B. Gavino, *Violence against women in Pakistan: A framework for analysis.* J Journal-Pakistan Medical Association, 2008. **58**(4): p. 198.
13. Khan, M.A., S.R. Jafree, and T. Jibeen, *Social Barriers to Mental Well-Being in Women of South Asia.* J The Sociology of South Asian Women's Health, 2020: p. 129-150.
14. Jalal, S. and M.Z. Younis, *Aging and elderly in Pakistan.* J Ageing International, 2014. **39**(1): p. 4-12.
15. NIPS, *Health Survey (PDHS)(2012-13)- National Institute of Population Studies,.* 2017.
16. Health Planning, S.S.a.I.A.U.H. and R.a.C. Ministry of National Health Services, *Pakistan Monitoring Report - Universal Health Coverage*. 2021.
17. Cheema, A.R., et al., *Availability Does Not Mean Utilisation: Analysis of a large Micro Health Insurance Programme in Pakistan.* J Global Journal of Health Science, 2020. **12**(10): p. 1-14.
18. Jalal, S., *The lady health worker program in Pakistan—a commentary.* J European journal of public health, 2011. **21**(2): p. 143-144.
19. Sarfraz, M. and S. Hamid, *Challenges in delivery of skilled maternal care–experiences of community midwives in Pakistan.* J BMC pregnancy childbirth, 2014. **14**(1): p. 1-13.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

20. Douthwaite, M. and P. Ward, *Increasing contraceptive use in rural Pakistan: an evaluation of the Lady Health Worker Programme.* J Health policy planning, 2005. **20**(2): p. 117-123.
21. Rahman, A., et al., *Cognitive behaviour therapy-based intervention by community health workers for mothers with depression and their infants in rural Pakistan: a cluster-randomised controlled trial.* J The Lancet, 2008. **372**(9642): p. 902-909.
22. Bhutta, Z.A., et al., *Implementing community-based perinatal care: results from a pilot study in rural Pakistan.* J Bulletin of the World Health Organization, 2008. **86**: p. 452-459.
23. Shaikh, B.T. and J. Hatcher, *Health seeking behaviour and health service utilization in Pakistan: challenging the policy makers.* J Journal of public health, 2005. **27**(1): p. 49-54.
24. Hasnain, S. and N. Sheikh, *Causes of low tetanus toxoid vaccination coverage in pregnant women in Lahore district, Pakistan.* %J EMHJ-Eastern Mediterranean Health Journal, 13 2007.
25. Organization, W.H., *Community health workers: a strategy to ensure access to primary health care services*. 2016: World Health Organization. Regional Office for the Eastern Mediterranean.
26. Punjab., G.o., *Minimum service delivery standards for primary and secondary health care in Punjab. Lahore: Government of Punjab, 2010.*
27. Hafeez, A., et al., *Lady health workers programme in Pakistan: challenges, achievements and the way forward.* J JPMA. The Journal of the Pakistan Medical Association, 2011. **61**(3): p. 210.
28. Jafree, S.R., et al., *Infection preparedness of community health workers: implications for maternal and neonatal health services in Pakistan.* J Primary Health Care Research Development, 2022. **23**.
29. Mumtaz, Z., et al., *Gender-based barriers to primary health care provision in Pakistan: the experience of female providers.* J Health policy planning, 2003. **18**(3): p. 261-269.
30. Panezai, S., M.M. Ahmed, and S.E. Saqib, *Gender differences in client satisfaction and its relationship with utilization of primary health care services in Pakistan.* J Journal of Geography Social Sciences, 2019. **1**(1): p. 30-43.
31. Jafree, S.R., et al., *Protocol for a systematic review of barriers, facilitators and outcomes in primary healthcare services for women in Pakistan.* J BMJ open, 2021. **11**(3): p. e043715.
32. University of Oxford, *Randomised Controlled Trials (RCT) Critical Appraisal Sheet.* 2021.
33. Hong, Q.N., et al., *The Mixed Methods Appraisal Tool (MMAT) version 2018 for information professionals and researchers.* J Education for information, 2018. **34**(4): p. 285-291.
34. Ali, B.S., et al., *The effectiveness of counseling on anxiety and depression by minimally trained counselors: a randomized controlled trial.* J American Journal of Psychotherapy, 2003. **57**(3): p. 324.
35. Ali, N.S., et al., *Effectiveness of counseling for anxiety and depression in mothers of children ages 0-30 months by community workers in Karachi, Pakistan: a quasi experimental study.* J BMC psychiatry, 2010. **10**(1): p. 1-9.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

36. Rahman, A., et al., *Effectiveness of a brief group psychological intervention for women in a post-conflict setting in Pakistan: a single-blind, cluster, randomised controlled trial.* J The Lancet, 2019. **393**(10182): p. 1733-1744.
37. Hackett, K., et al., *Impact of home-based family planning counselling and referral on modern contraceptive use in Karachi, Pakistan: a retrospective, cross-sectional matched control study.* J BMJ open, 2020. **10**(9): p. e039835.
38. Hameed, W., et al., *Comparing effectiveness of active and passive client follow-up approaches in sustaining the continued use of long acting reversible contraceptives (LARC) in Rural Punjab: a multicentre, non-inferiority trial.* J PloS one, 2016. **11**(9): p. e0160683.
39. Sikander, S., et al., *Delivering the Thinking Healthy Programme for perinatal depression through volunteer peers: a cluster randomised controlled trial in Pakistan.* J The Lancet Psychiatry, 2019. **6**(2): p. 128-139.
40. Sikander, S., et al., *Cognitive-behavioral counseling for exclusive breastfeeding in rural pediatrics: a cluster RCT.* J Pediatrics, 2015. **135**(2): p. e424-e431.
41. Khan, M., et al., *Evaluating feasibility and acceptability of a group WHO trans-diagnostic intervention for women with common mental disorders in rural Pakistan: a cluster randomised controlled feasibility trial.* J Epidemiology psychiatric sciences, 2017. **28**(1): p. 77-87.
42. Kumar, R., et al., *Effectiveness of a health education intervention on the use of long-lasting insecticidal nets for the prevention of malaria in pregnant women of Pakistan: a quasi-experimental study.* J Malaria Journal, 2020. **19**(1): p. 1-10.
43. Shallwani, K., et al., *Self-examination for breast and testicular cancers: a community-based intervention study.* J Asian Pacific journal of cancer prevention: APJCP, 2010. **11**(1): p. 145.
44. Jokhio, A.H., H.R. Winter, and K.K. Cheng, *An intervention involving traditional birth attendants and perinatal and maternal mortality in Pakistan.* J New England Journal of Medicine, 2005. **352**(20): p. 2091-2099.
45. Omer, K., et al., *Evidence-based training of frontline health workers for door-to-door health promotion: a pilot randomized controlled cluster trial with Lady Health Workers in Sindh Province, Pakistan.* J Patient education counseling, 2008. **72**(2): p. 178-185.
46. Hirani, S.S., et al., *Social support intervention to promote resilience and quality of life in women living in Karachi, Pakistan: a randomized controlled trial.* J International journal of public health, 2018. **63**(6): p. 693-702.
47. Khushk, W.A., et al., *Health and social impacts of improved stoves on rural women: a pilot intervention in Sindh, Pakistan.* J Indoor air, 2005. **15**(5): p. 311-316.
48. Maselko, J., et al., *Effectiveness of a peer-delivered, psychosocial intervention on maternal depression and child development at 3 years postnatal: a cluster randomised trial in Pakistan.* J The Lancet Psychiatry, 2020. **7**(9): p. 775-787.
49. Agha, S., *Impact of a maternal health voucher scheme on institutional delivery among low income women in Pakistan.* J Reproductive health, 2011. **8**(1): p. 1-14.
50. Azmat, S.K., et al., *Impact of social franchising on contraceptive use when complemented by vouchers: a quasi-experimental study in rural Pakistan.* J PLoS One, 2013. **8**(9): p. e74260.
51. Azmat, S.K., et al., *Engaging with community-based public and private mid-level providers for promoting the use of modern contraceptive methods in rural Pakistan:*

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

results from two innovative birth spacing interventions. J Reproductive health, 2016. **13**(1): p. 1-15.
52. Najmi, H., et al., *Community-based integrated approach to changing women's family planning behaviour in Pakistan, 2014–2016.* J Public health action, 2018. **8**(2): p. 85-90.
53. Midhet, F. and S. Becker, *Impact of community-based interventions on maternal and neonatal health indicators: Results from a community randomized trial in rural Balochistan, Pakistan.* J Reproductive health, 2010. **7**(1): p. 1-10.
54. Qureshi, R.N., et al., *Community-Level Interventions for Pre-eclampsia (CLIP) in Pakistan: a cluster randomised controlled trial.* J Pregnancy hypertension, 2020. **22**: p. 109-118.
55. Ali, M., et al., *Assessing Effectiveness of Multipurpose Voucher Scheme to Enhance Contraceptive Choices, Equity, and Child Immunization Coverage: Results of an Interventional Study from Pakistan.* J Journal of Multidisciplinary Healthcare, 2020. **13**: p. 1061.
56. Mirza, I., et al., *Community management of intellectual disabilities in Pakistan: a mixed methods study.* J Journal of Intellectual Disability Research, 2009. **53**(6): p. 559-570.
57. Singh, K., et al., *Multimorbidity in South Asian adults: prevalence, risk factors and mortality.* J Journal of Public Health, 2019. **41**(1): p. 80-89.
58. Zakar, R., Zakar, M. Z., & Kraemer, A. , *Primary health care physicians' response to the victims of spousal violence against women in Pakistan.* . Health care for women international, 32(9), 811-832., 2011.
59. Habumuremyi P D, Z.M., *"Making Family Planning a National Development Priority."* The Lancet 380 (9837): 78–80, 2012.
60. Bruce, J., *Fundamental elements of the quality of care: a simple framework.* J Studies in family planning, 1990. **21**(2): p. 61-91.
61. Sheikh, K.S., K. Ahmad, and A. Farooq, *Socio-Political and Cultural Challenges of Poverty Alleviation: A Comparative Analysis of Pakistan and Bangladesh.* J South Asian Studies, 2016. **31**(2): p. 291.
62. Malik, M.N., M.S. Awan, and T. Saleem, *Social mobilization campaign to tackle immunization hesitancy in Sargodha and Khushab districts of Pakistan.* J Journal of Global Health, 2020. **10**(2).
63. Naqvi, Z.Z. and S. Riaz, *Women in Pakistan: Countering conflicts and building peace.* J Asian Journal of Women's Studies, 2015. **21**(3): p. 326-338.
64. Omer, S., et al., *The influence of social and cultural practices on maternal mortality: a qualitative study from South Punjab, Pakistan.* J Reproductive health, 2021. **18**(1): p. 1-12.
65. Mumtaz, Z. and S. Salway, *'I never go anywhere': extricating the links between women's mobility and uptake of reproductive health services in Pakistan.* J Social science medicine, 2005. **60**(8): p. 1751-1765.
66. Zaidi, S., et al., *Expanding access to healthcare in South Asia.* J BMJ open, 2017. **357**.
67. D'Zurilla, T.J. and A.M. Nezu, *Problem-solving therapy.* J Handbook of cognitive-behavioral therapies, 2010. **3**: p. 197-225.
68. Di Cesare, M., et al., *Geographical and socioeconomic inequalities in women and children's nutritional status in Pakistan in 2011: an analysis of data from a nationally representative survey.* J The Lancet Global Health, 2015. **3**(4): p. e229-e239.

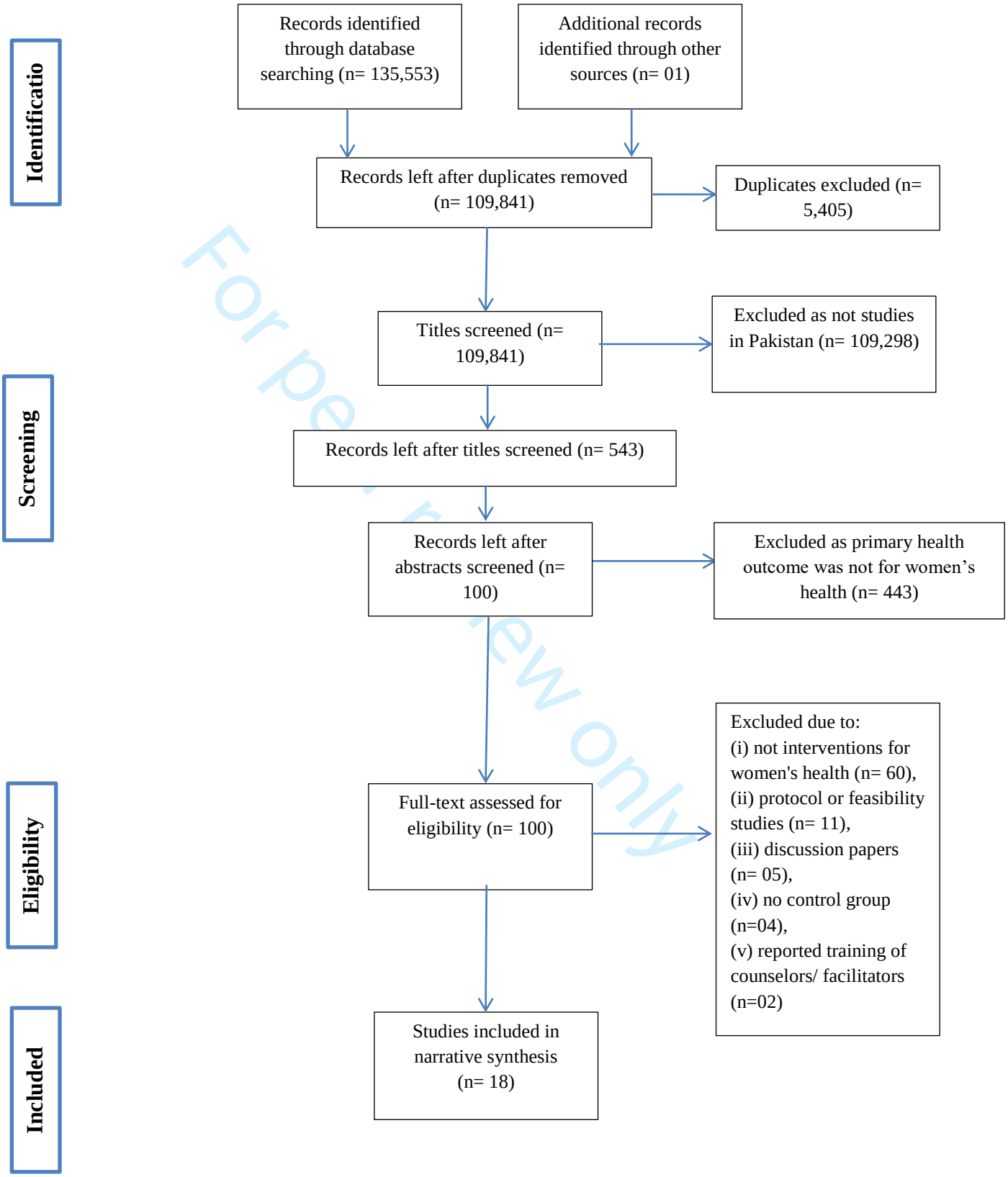
For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

69. Ahmed, S., et al., *Autonomous decision-making for antenatal screening in Pakistan: views held by women, men and health professionals in a low–middle income country.* J European Journal of Human Genetics, 2019. **27**(6): p. 848-856.
70. Jahagirdar, D., et al., *Using patient reported outcome measures in health services: a qualitative study on including people with low literacy skills and learning disabilities.* J BMC health services research, 2012. **12**(1): p. 1-8.
71. Meadows, K.A., *Patient-reported outcome measures: an overview.* J British journal of community nursing, 2011. **16**(3): p. 146-151.
72. Muhammad Gadit, A.A. and G. Mugford, *Prevalence of depression among households in three capital cities of Pakistan: need to revise the mental health policy.* J Plos one, 2007. **2**(2): p. e209.
73. Londoño, J.-L. and J. Frenk, *Structured pluralism: towards an innovative model for health system reform in Latin America.* J Health Policy, 1997. **41**(1): p. 1-36.

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Figure 1-
Preferred Reporting Items for Systematic Reviews and Meta-Analyses flow chart



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Supplementary File 1-

Search Terms and Database Results

Search terms

1. primary health OR community health AND Pakistan AND women

2. primary healthcare OR primary health care OR primary service delivery OR primary health service* OR primary health evaluation OR program* OR polic* AND Pakistan AND women

3. primary health OR community health OR wom#n OR mother OR maternal OR women's health OR maternal health OR reproductive health AND Pakistan AND women

4. primary health OR community health OR evidence based practice OR intervention OR RCTs OR randomi*ed controlled trial* OR cluster randomi*ed controlled trial* OR clinical trial* OR quasi-experimental OR non-randomi*ed OR cross-sectional OR mixed methods OR quantitative OR qualitative OR case-study OR group studies OR evidence based OR cohort studies OR case-control studies AND Pakistan AND women

5. primary health OR community health OR (lady health worker OR LHW) OR (community health worker OR CHW) OR (community midwife OR CMW) OR (traditional birth attendant OR TBA) AND Pakistan AND women

Table:
Summary of results from databases searched, as at August ….

Search Term	PubMed	BMC	Medline	CINAHL	Cochrane Library	Total
1.	4,074	2,435	207	1,756	1,679	10,151
2.	1,702	1,769	8,088	2,522	2,438	16,519
3.	4,427	2,435	207	2,686	2,597	12,352
4.	5,029	2,436	23,894	7,819	7,688	46,866
5.	5,458	595	52	0	8,340	14,445
6.	5,340	2,435	23,665	1,931	1,849	35,220
	26,030	**12,105**	**56,113**	**16,714**	**24,591**	**135,553**

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Supplementary File 2: Quality Appraisal of Included Studies

A. RCT Critical Appraisal

Table Key:

Initial critical appraisal screening:

1. Does this study address a clearly focused question?
2. Did the study use valid methods to address this question?
3. Are the valid results of this study important?
4. Are these valid, important results applicable to my patient or population?

The RCT Critical Appraisal Checklist screening questions:

5. Was the assignment of patients to treatments randomized?
6. Were the groups similar at the start of the trial?
7. Aside from the allocated treatment, were groups treated equally?
8. Were all patients who entered the trial accounted for? And were they analysed in the groups to which they were randomised?
9. Were measures objective or were the patients and clinicians kept "blind" to which treatment was being received?
10. Are there suitable measures used to report results and are the estimates of the treatment effect precise?

Table 1: Critical assessment of included articles (RCTs)

No.	First author (Year)	Type of Study	1	2	3	4	5	6	7	8	9	10	Score
1	Ali, B. S (2003)	RCT	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	High
2	Khan, M. N. (2017)	Cluster RCT	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	High
3	Rahman, A (2019)	Cluster RCT	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	High
5	Sikander, S. (2019)	Cluster RCT	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	High
6	Rahman, A (2008).	RCT	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	High
8	Sikander, S. (2015)	RCT	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	High
9	Jokhio, A. H (2005)	Cluster RCT	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	High
10	Omer, K. (2008)	Cluster RCT	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	High
12	Hirani, S. (2018)	RCT	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	High
13	Maselko, J (2020)	RCT	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	High
17	Midhet, F (2010)	RCT	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	High
18	Qureshi, R. N (2020)	RCT	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	High

Scoring Key: Each assessment criterion was scored (Yes, No, Unclear)

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B. MMAT appraisal

Checklist for included studies which fit the criterion, as described in MMAT appraisal as: Quantitative non-randomized studies (Non-randomized studies are defined as any quantitative studies estimating the effectiveness of an intervention or studying other exposures that do not use randomization to allocate units)

Screening Questions
1. Are there clear research questions?
2. Do the collected data allow to address the research questions?

Methodological quality criteria
3. Are the participant's representative of the target population?
4. Are measurements appropriate regarding both the outcome and intervention (or exposure)?
5. Are there complete outcome data?
6. Are the confounders accounted for in the design and analysis?
7. During the study period, is the intervention administered (or exposure occurred) as intended?

Table 2: Critical assessment of included articles (non-randomized studies)

No.	First author (Year)	Type of Study	1	2	3	4	5	6	7	Score	Comments
1	Ali, N. S (2010)	Non-randomized quasi-experiment	Yes	Yes	Yes	Yes	Yes	No	No	Medium	This was a non-randomized sample from an action research program.
2	Hameed, W (2016)	Non-randomized study non-inferiority trial	Yes	Yes	Yes	Yes	Yes	Yes	No	Medium	Union council areas were sampled randomly. Participants & could not be randomized, (due to the low proportion of LARC users with access to phones)
3	Kumar, R. (2020)	Non-randomized quasi-experiment	Yes	Yes	Yes	Yes	Yes	Yes	No	Medium	Union council areas were sampled randomly. However, women were sampled in a non-randomized manner (as many pregnant women did not consent to participate due to cultural or health issues).
4	Azmat, S. K. (2013)	Non-randomized quasi-experiment	Yes	Yes	Yes	Yes	Yes	Yes	No	Medium	The control sample could not be randomly selected.
5	Azmat, S. K. (2016)	Non-randomized quasi-experiment	Yes	Yes	Yes	Yes	Yes	Yes	No	Medium	Non-randomized sample (as two intervention models were being compared- community midwife model vs. voucher scheme)
6	Ali, M (2020)	Non-randomized quasi-experiment	Yes	Yes	Yes	Yes	Yes	Yes	No	Medium	Union council areas were sampled randomly. Participants were not sampled randomly, but passed on willingness to participate.

Scoring Key: Each assessment criterion was scored (Yes, No, Unclear)

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Supplementary File 3:

Characteristics of Included Studies

Table 1:
Included intervention-based studies

No.	First author	Year	Study Design	Setting	Participants	Sample size	Aim	Type of Intervention	Outcome being measured post intervention	Intervention Delivered by
Counseling Interventions										
1	Ali, B. S	2003	RCT	Semi-urban community in Karachi city	Women aged 18 to 50 years, screened for anxiety and/or depression	Experiment: n= 70 Control: n= 91	Assess if counseling sessions reduce anxiety and/or depression	Mental Health Counseling (one on one)	Reduction in the mean anxiety and depression scores	Briefly trained women counselors of community
2	Khan, M. N.	2017	Cluster RCT	Swat rural district	Women aged 18 or more years affected by conflict	Experiment: n= 59 Control: n= 60	Evaluate the feasibility and acceptability of a group psychotherapeutic interventions, based upon cognitive behavioral approach, to reduce anxiety and depression	Mental Health Counseling (in group)	Decline in psychological distress, measured by levels of anxiety and depression, post-traumatic stress disorder (PTSD), and assessment of general psychological profile and levels of functioning	Briefly trained female non-specialists and LHWs
3	Rahman, A	2019	Cluster RCT	Two rural union councils of Swat city (Odigram and Ghalegay)	Women aged 18–60 years affected by conflict	Experiment: n= 288 Control: n= 290	Effectiveness of a group psychological intervention to improve mental health	Mental Health Counseling (group session)	Reductions in symptoms of anxiety, depression, PTSD, and functional impairment, improved social support, and reduced prevalence of depressive disorders 3 months after the intervention	Briefly trained female non-specialists and LHWs
4	Ali, N. S	2010	Non-randomized quasi-experiment	Qayoomabad and Manzoor Colony of Karachi city	Mothers (with children 0-30 months) screened for postnatal depression	Experiment: n= 62 Control: n= 12	Assess the benefits of counseling on reducing postnatal depression	Mental Health Counseling (one on one)	Reduction in the mean anxiety and depression scores	Women from the community
5	Sikander, S.	2019	Cluster RCT	40 village clusters in Rawalpindi	Pregnant women	Experiment: n= 283 Control: n= 287	Aimed to reduce perinatal depression	Mental Health Counseling (one on one and group)	Severity of depressive symptoms and the prevalence of remission 6 months after childbirth	Volunteer peers and LHWs

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6	Rahman, A.	2008	RCT	40 Union Council clusters in rural Rawalpindi city	Pregnant women	Experiment: n= 463 Control: n= 440	Provide a CBT intervention for mothers with depression	Mental Health Counseling (group session)	Infant weight and height at 6 months and 12 months and prevalence of maternal depression	Trained LHWs
7	Hameed, W	2016	Non-inferiority trial without randomisation	Twenty-two (16 rural- and 6 urban-based) franchised reproductive healthcare facilities in district Chakwal	Married women using long-acting reversible contraceptive	Experiment: n= 1,246 Control: none	Compare the effectiveness of active (doorstep and telephonic) and passive (needs-based response) follow-up (counseling) in sustaining the use of long-acting reversible contraceptive (LARC)	Family Planning Counseling (one on one)	Sustaining the use of long-acting reversible contraceptive	Female Health Educators
8	Sikander, S.	2015	RCT	40 Union Councils in Manshera city	Pregnant women	Experiment: n= 210 Control: n= 211	Assess the effectiveness of cognitive-behavioral counseling on the rate and duration of exclusive breastfeeding (EBF) during the first 6 months of an infant's life	Family Planning Counseling (one on one)	Increase in exclusive breastfeeding and decline in use prelacteal feeds	LHWs
Health Education and Awareness Interventions										
1	Jokhio, A. H	2005	Cluster RCT	7 subdistricts Larkana city, Sindh province	Pregnant women	Experiment: n= 10,093 Control: n= 9,432	Provide training and other support to TBAs to reduce perinatal and maternal mortality and reduce complications of pregnancy	Health education and awareness	Decline in perinatal death and maternal mortality	TBAS, LHWs and Obstetricians
2	Omer, K.	2008	Cluster RCT	10 underdeveloped areas from three cities in Sindh (Karachi West, Karachi South and Hyderabad)	Pregnant women	Experiment: n= 529 Control: n= 541	Assess the impact of developing community-based communication tools to promote favorable maternal health practices	Health education and awareness	Improvement in prenatal checkups, care during pregnancy, feeding baby colostrum and maintaining exclusive breastfeeding	LHWs
3	Kumar, R.	2020	Non-randomized quasi-experiment	Two Union Councils (UCs) of Tarparkar district, Sindh	Pregnant women with children up to 6 months of age	Experiment: n= 100 Control: n= 100	Effectiveness of a health education intervention for the prevention of malaria in pregnant women with children up to 6 months of age	Health education and awareness	Knowledge of malaria prevention & use of long-lasting insecticide treated bed nets	LHWs

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				Province						
Social and Psychosocial Interventions										
1	Hirani, S. S.	2018	RCT	Low socioeconomic areas of Karachi city	Young adult women, 20–45 years	Experiment: n= 60 Control: n= 60	Provide a social support intervention to enhance resilience and quality of life	Social, psychosocial or preventive *Social- group sessions*	Improvement in resilience and quality of life	CHWs
2	Maselko, J	2020	RCT	40 village clusters in Kallar Syedan, a rural subdistrict of Rawalpindi city	Pregnant women	Experiment: n= 206 Control: n= 683	Assess the effectiveness of psychosocial intervention on maternal depression and child development (group session)	Social, psychosocial or preventive *Psychosocial-group sessions*	Decline in maternal depression symptoms and remission	Non-specialists peers (lay married women who lived in the same community as that of the women with depression and volunteered their time)
Combination Interventions										
1	Azmat, S. K.	2013	Non-randomized quasi-experiment	Rural and under-served communities of (i) Badin and Dadu-Sindh province, and (ii) Jhang and Khanewal-Punjab province	Married women reproductive age	Experiment: n= 2,483 Control: n= 2,509	Evaluate the effectiveness of a social franchise programme along with a free voucher scheme to promote awareness and use of modern long term contraceptive	Combination	Assess improvement in contraceptive awareness and long-term contraception use	Female health visitors, midwives and nurses at their respective private clinics
2	Azmat, S. K.	2016	Non-randomized quasi-experiment	Eight districts of Sindh, Punjab and KPK	Married women reproductive age	Experiment: n= 1,817 Control: n= 1,075	Provide accessible and affordable long term family planning services through marketing, branding and introducing a voucher scheme	Combination	Increasing awareness about FP methods and use of contraception	Trained facilitators and CMWs
3	Ali, M	2020	Non-randomized quasi-experiment	Faisalabad (urban) & Toba Tek Singh (rural),	Married women of reproductive age	Experiment: n= 1,276 Control: n= 1,276	Assess the effectiveness of a subsidized, multipurpose voucher intervention to enhance the client–provider interaction for improved	Combination	Increase in modern contraception use, knowledge of contraceptives, receiving ANC, and delivery at health	Vouchers and awareness outreach through service providers and distributing

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				Punjab			contraceptive counseling, modern contraception methods uptake, continued use, and its impact on equity through better targeting, while increasing uptake of postnatal care and child immunization among women from the lowest two wealth quintiles.		facilities (also child vaccination rates for BCG, DPT, HBV, and measles)	agencies
4	Midhet, F	2010	RCT	32 village clusters in Khuzdar, a rural district of Baluchistan province	Pregnant women	Experiment: n= 1,539 Control: n= 1,022	Assess the impact of education sessions for safe motherhood practices, TBA training, and emergency transport system on maternal and neonatal health indicators	Combination	Reduction of delays in seeking medical care for obstetric complications and reduction in perinatal and neonatal mortality	Local women with secondary school education or above, & TBAs
5	Qureshi, R. N	2020	RCT	Matiari and Hyderabad districts, Sindh Province	Pregnant women screened for preeclampsia	Experiment: n= 20,264 Control: n= 19,182	Assess the impact of combined interventions for pre-eclampsia (Mobile health assessment; referral; doorstep visits; and educational sessions)to reduce all-cause maternal and perinatal mortality and major morbidity	Combination	Reduction in stillbirths, maternal death or morbidity, early or late neonatal deaths and neonatal morbidity	LHWs

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Page 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Page 2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Page 3-5
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Page 3-5
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Page 6-7
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Page 6
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Page 6 (+SuppFile1)
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Page 6-7
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Page 7-8
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Page 7-8
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Page 7-8
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Page 8
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	n/a
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Page 9
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Page 9
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Page 9
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Page 9
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	n/a
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	n/a
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	n/a

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Page 8
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Page 9 (Fig 1)
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Page 7
Study characteristics	17	Cite each included study and present its characteristics.	Page 12-13 (Table 1)
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	n/a
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Page 14-21
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Page 14-21
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Page 14-21
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Page 14-21
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	n/a
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	n/a
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Page 8
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Page 22-26
	23b	Discuss any limitations of the evidence included in the review.	Page 25-26
	23c	Discuss any limitations of the review processes used.	Page 25-26
	23d	Discuss implications of the results for practice, policy, and future research.	Page 26
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Page 2
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Page 2
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	n/a
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page 27
Competing interests	26	Declare any competing interests of review authors.	Page 27
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Page 27

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PRISMA 2020 Checklist

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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